Cross-country heterogeneity in potential-GDP and NAIRU hysteresis: A Bayesian approach*

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Abstract

This paper extends the semi-structural New Keynesian framework to estimate the potential GDP hysteresis elasticity (η) and the NAIRU hysteresis elasticity (η_u) jointly, by Bayesian methods, for 30 economies. The cross-country averages are 0.46 for η and 0.40 for η_u, implying that 0.63 of a temporary demand shock remains in GDP after ten years. Regarding η_u, its cross-country variation is high (0.04 in Iceland and 0.73 in Portugal) and seems to be higher in economies with stricter employment protection laws and in economies whose monetary policy spent longer at its lower bound. Cross-country heterogeneity of η is smaller in statistical terms but economically material (0.25 in Korea and 0.71 in Spain), and none of the structural characteristics considered accounts for it. Allowing for positive hysteresis elasticities yields measured output gaps that are smaller and less persistent. Using the identified demand shocks, local projections show that employment and capital respond persistently to demand shocks while measured TFP does not, a pattern consistent with accumulation-based mechanisms rather than TFP channels. These findings imply that the costs of recessions and aggressive disinflation are substantially higher than conventional frameworks suggest.

Keywords: hysteresis, business cycle, Bayesian methods, Kalman filter, potential GDP, NAIRU

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1 Introduction

Modern mainstream macroeconomics has been highly influenced by Milton Friedman's natural rate hypothesis. The natural rate hypothesis posited by Friedman (1968) is composed of two statements: the independence hypothesis and the accelerationist hypothesis. The former suggests that monetary policy cannot permanently affect the unemployment rate or GDP, and the latter suggests that if central banks try to increase GDP permanently, they will face persistently higher inflation. Both statements shape the way we think about the economy. It is canon to assume that variables like potential GDP and the NAIRU are independent of the economic cycle and monetary policy. This independence hypothesis has been contested by the hysteresis hypothesis, which suggests that potential GDP and the NAIRU can be affected by short-run fluctuations (see Cerra et al. (2023) for an overview).

Testing the independence hypothesis requires estimating its key parameters. To fix ideas, consider a simplified framework where the evolution of potential GDP ($\bar{y}_t$) and the NAIRU ($\bar{u}_t$) is governed by the following processes:

$$\begin{aligned} \bar{y}_t &= \bar{y}_{t-1} + \eta\, \tilde{y}_{t-1} + \text{trend shocks}_t, \\ \bar{u}_t &= \bar{u}_{t-1} + \eta_u\, \tilde{u}_{t-1} + \text{NAIRU shocks}_t. \end{aligned} \tag{1}$$

where $\tilde{y}_{t-1}$ and $\tilde{u}_{t-1}$ are the past output and unemployment gaps, respectively. η is the potential output hysteresis elasticity and η_u is the NAIRU hysteresis elasticity. The independence assumption in its strict sense implies that $\eta = 0$ and $\eta_u = 0$, while in a broader sense, it implies that these parameters are small (see Blanchard, 2018). In contrast, the hysteresis hypothesis means that these parameters are high.[1] Despite their centrality to this debate, country-specific estimates of both η and η_u across a broad sample of economies do not exist. This paper provides them.

The empirical challenge is that η and η_u relate latent variables, i.e. the output gap with potential output, and the unemployment gap with the NAIRU, respectively, so the

[1] It is important to stress that hysteresis literature has, until recently, relied on unit-root tests to assess whether the natural rate hypothesis holds. However, in this simple model shown in equation (1), it can be seen that the independence hypothesis might hold even if the variables have unit-roots. The rationale is that there could exist technological shocks that affect potential GDP permanently or labour reforms that affect the NAIRU permanently. The hysteresis hypothesis suggests that other factors such as demand shocks or monetary policy shocks could affect permanently GDP or the unemployment rate. This could only happen in this simple model if $\eta>0$ and/or $\eta_u>0$.

estimation of these parameters and the latent variables should be done simultaneously. Thus, I extend the class of semi-structural New Keynesian models of Blagrave et al. (2015) and Holston et al. (2017) to treat η and η_u as free parameters estimated by Bayesian methods. The model is composed of four blocks. GDP is modeled with an IS curve and a law of motion for potential output. The unemployment rate follows a dynamic Okun's law and a law of motion for the NAIRU. Headline inflation follows a Phillips curve and the short term interest rate follows a Taylor rule. I apply the framework, country by country, to 30 economies. Furthermore, using panel local projections on the model's demand shocks, I inspect how demand feeds back into supply through employment, capital, and TFP. Finally, I use meta-analytic and model-averaging tools to explore the cross-country heterogeneity on η and η_u. From all these exercises, I found four main results.

First, high levels of hysteresis seem to be the norm. The cross-country average for η is 0.46[2]: roughly forty-six per cent of the prevailing output gap is absorbed into potential output each period. Cumulated, this means that a demand shock raising GDP by one percent still leaves an effect of 0.63 percent at the ten-year horizon, and in some economies the long-run effect exceeds the initial impulse. A 100 basis point monetary policy tightening is followed by a persistent GDP shortfall of about 0.64 per cent. The average NAIRU elasticity η_u is 0.40, so cyclical unemployment leaves a highly persistent — though, with $\rho_u = 0.95$, ultimately mean-reverting — imprint on structural unemployment.

Second, hysteresis varies across economies, and by economically meaningful amounts. Using meta-analytic techniques on the estimates, I find the dispersion in η_u to be larger than in η, but the latter is far from negligible. As a measure of this dispersion, the 95% prediction interval for the value of η in an economy outside the sample runs from 0.28 to 0.64, and for η_u from 0 to 0.83. Those ranges have different implications. Carried through to the responses they generate, the ten-year GDP effect of a demand shock lies between 0.34 and 0.92, and the five-year unemployment effect of a labour demand shock between 0 and 1.25. In addition, both potential output and NAIRU hysteresis are higher in Europe ($\eta = 0.50$ and $\eta_u = 0.48$, on average) than in the rest of the sample ($\eta = 0.38$ and $\eta_u = 0.24$, on average), aligned with stylized facts documented by Blanchard and Summers (1986) and Galí (2022), both of which highlight Europe specifically. Finally,

[2]All cross-country averages reported in the paper are the random-effects pooled means obtained from the meta-analysis of the country-specific posteriors in Section 6, which weight each economy by the precision of its estimate, rather than simple arithmetic means. The corresponding intervals and heterogeneity statistics are reported there.

an exploratory analysis of which structural features of the economy correlate with these estimates finds η_u to be higher in countries with stricter employment protection laws and in countries whose monetary policy spent longer at its lower bound, while for η no characteristic is robustly associated with its size.

The third finding relates to the characteristics of potential output and NAIRU under hysteresis. Allowing for $\eta > 0$ and $\eta_u > 0$ changes the output gap and unemployment gap estimates making them less volatile and less persistent, and the trend components less smooth, than their estimates under the natural rate hypothesis ($\eta = 0$ and $\eta_u = 0$). I found a similar pattern when analyzing the estimates published by the IMF and the OECD, which record a pre-2008 boom almost as large as the contraction that followed while the estimates here do not. The trade-off between a smooth trend and a large, persistent cycle is familiar from the trend-cycle literature (see Morley et al. (2003)), but it arrives here through an estimated mechanism rather than a covariance restriction.

Finally, the paper's last finding is about the mechanisms through which demand shocks transmit to the supply side of the economy. Using the identified demand shocks from the model, I estimated the impact on GDP, TFP, employment and capital of demand shocks. Employment and the capital stock respond persistently while TFP effects are transitory. A sanity check using supply shocks yields the expected pattern: permanent GDP increases driven by TFP and capital, with no significant long-run effect on employment. This result is informative about the sources of hysteresis in GDP.

The paper contributes to the literature on potential output and NAIRU estimation in the semi-structural tradition of Blagrave et al. (2015) and Holston et al. (2017). Most official and academic estimates in that tradition do not allow for hysteresis; treating η and η_u as free parameters relaxes this assumption and lets the data speak. Existing estimates of η (Jordà et al., 2020; Roa Rozo, 2024) are pooled cross-country averages, and the country-specific estimates of η_u (Calvert Jump and Stockhammer, 2019, 2022, 2023; Clavijo-Cortes, 2023) cover only selected European economies or Latin America. This is the first paper to estimate both elasticities jointly, for thirty economies, on one method. Two papers in this tradition allow for hysteresis, Alichi et al. (2019) for the United States and Ramos-Veloza et al. (2026) for Colombia. In both hysteresis affects the NAIRU and potential output is affected by those changes in the NAIRU. Thus, output hysteresis that does not run through unemployment cannot be seen; while in this paper both channels

are free.

The paper also contributes to the literature on hysteresis mechanisms. Cerra et al. (2023) identify three frameworks that generate hysteresis in GDP: learning-by-doing (Lucas, 1988), AK models (Romer, 1986) and endogenous R&D (Romer, 1990). The local projections help to distinguish among them. Employment and the capital stock respond persistently to demand shocks, while the response of measured TFP is transitory. This is consistent with mechanisms that work through the accumulation of capital and labour, and less so with endogenous R&D, which works through technology. Regarding unemployment hysteresis, the cross-country evidence is suggestive rather than conclusive. η_u is higher in economies with stricter employment protection and in economies whose monetary policy spent longer at its lower bound and, among advanced economies, more weakly in those with an older population. It is not associated with the labour share, union density or bargaining coverage, the usual proxies for insider power. Thus, the results favour models based on firing costs and skill loss (Bentolila and Bertola, 1990; Ljungqvist and Sargent, 1998), on a constrained policy response (Benigno and Fornaro, 2018; Garga and Singh, 2021), and more weakly on the re-employment prospects of older workers (Chan and Stevens, 2001; Neumark et al., 2019), over the insider–outsider tradition that has organised the literature since Blanchard and Summers (1986), at least with the proxies used here.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 presents the model and the elicitation of the prior distributions. Section 4 reports the estimates of η and η_u, the long-run effects of demand and monetary policy shocks, and the estimates of potential output and the NAIRU they imply. Section 5 uses the shocks identified by the model to study the mechanisms behind hysteresis. Section 6 measures the cross-country heterogeneity in η and η_u and the structural characteristics associated with it. Section 7 concludes.

2 Data sources and sample

I used the Global Macro Database (GMD) collected by Müller et al. (2025). GMD is an open-source, continuously updated dataset of macroeconomic statistics that unifies and extends existing resources. By harmonising and integrating data from 32 major contem-

porary sources—including the IMF, World Bank, and OECD— with historical records from 78 additional datasets, Müller et al. (2025) constructed comprehensive annual time series for 46 variables across 243 countries. I used data from this dataset on real GDP, CPI, short-term interest rates, and unemployment rates.

The sample is comprised of the following economies: Australia, Austria, Belgium, Canada, Switzerland, Chile, China, Colombia, Germany, Denmark, Spain, Finland, France, the United Kingdom, Greece, Hungary, Ireland, Iceland, Israel, Italy, Japan, South Korea, Luxembourg, the Netherlands, Norway, New Zealand, Portugal, Sweden, the United States, and South Africa. This sample was chosen due to data availability and the absence of hyperinflation episodes after 1990. When available, data from 1950 onwards was used. In countries with hyperinflation episodes prior to 1990, data after 1990 were used. Figure A1 (in Appendix A) shows the time coverage for each country. This sample covered about 60% of global GDP in 2015.

To assess which mechanisms generate hysteresis effects, I use data from the Penn World Table version 11.0 (Feenstra et al., 2015). Specifically, I use total factor productivity at constant national prices, the capital stock at constant national prices, and employment measured as the number of persons engaged. All three series enter the local projections in logs. For the thirty economies in the sample, capital and employment are available from 1950 and TFP from the mid-1950s (with Hungary only from 1974), so for the economies whose GMD series start by then the overlap is essentially complete from the mid-1950s onward, with TFP the binding series; for those that enter the GMD sample later (Figure A1), the start of the model sample binds instead. The resulting panel is unbalanced.

The cross-country exercise of Section 6 relates the estimated elasticities to fifteen structural characteristics measured within 1980–2019. Nine are averages of annual series from the Penn World Table (Feenstra et al., 2015) and the World Development Indicators (World Bank, 2026), each taken over the longest span of at least ten years common to all economies, so that differences in country means reflect the economies and not the periods in which each was observed. The other six lack series long enough for that rule. Five enter as cross-sections: product market regulation (OECD, 2024), union density (2010–2019, from the OECD and academic research (OECD, 2026b; Qi and Huang, 2016)), collective bargaining coverage (ILO, 2026), R&D intensity, and the share of years in which the

policy rate stood at or below 0.50% for at least six months, which I construct from BIS data (BIS, 2026). The sixth, employment protection (OECD, 2026a), is a ten-year average for most economies but rests on five years of data for China and South Africa and on six for Colombia. Their coverage is uneven, so a result resting on them is weaker. Table H1 in Appendix H reports the source, window and coverage of each.

3 The model

The model here proposed is rather standard in the New Keynesian semi-structural models literature (see Blagrave et al. (2015) and Holston et al. (2017)). The main difference from the textbook version is that it allows for hysteresis effects. These differences are highlighted in blue. The model is composed of four blocks related to the four key variables in the model: GDP, the unemployment rate, inflation, and the monetary policy rate.

GDP growth dynamics are governed by the following 4 equations.

$$\Delta y_t = \Delta \bar{y}_t + \Delta \tilde{y}_t \tag{2}$$

$$\tilde{y}_t = \alpha\, \tilde{y}_{t-1} - \alpha_r \left(r_t - \bar{r}_t \right) - \alpha_s\, \varepsilon_t^{\bar{y}} + \varepsilon_t^{\tilde{y}} \tag{3}$$

$$\Delta \bar{y}_t = \gamma_t + \eta\, \tilde{y}_{t-1} + \varepsilon_t^{\bar{y}} \tag{4}$$

$$\gamma_t = \delta\, \gamma_{t-1} + (1-\delta)\, \gamma_{ss} + \varepsilon_t^{\gamma} \tag{5}$$

The GDP log difference Δy_t is divided into potential output growth $\Delta \bar{y}_t$ plus the change in the output gap $\Delta \tilde{y}_t$. The output gap $\tilde{y}_t$ follows an IS curve and the error term $\varepsilon_t^{\tilde{y}}$ is meant to capture all shocks on the right-hand side of the equation that can be thought of as demand shocks. The parameter α_r is the slope of the IS curve (where $r_t = i_t - \pi_{t+1|t}$) and the parameter α_s measures the response of the output gap to a supply (trend) shock. Potential GDP is a unit-root process with a drift γ_t, which is the trend (exogenous) component of potential GDP growth. γ_t is a stationary process over a constant long-run growth rate γ_{ss}.

The main difference in contrast to an otherwise standard New Keynesian model is found in equation (4). In most models, it is assumed that $\eta = 0$, while in this study this parameter is estimated. The specification found in equation (4) is closely related to Jordà et al. (2020) who suggest TFP (Z_t) follows the process: $Z_t = Z_{t-1} + \mu + \eta\, \tilde{y}_{t-1}$,

where μ is the long-run TFP growth (it can be subject to trend shocks ε_t^{γ}). In their model, TFP growth is equal to potential GDP growth, so this reduced form is valid. It is also related to DeLong and Summers (2012) who defined hysteresis elasticity as the percent reduction in the flow of future potential output per percentage-point-year of the present-period output gap. Finally, Cerra et al. (2023) also suggest this reduced form as the simplest way to allow for hysteresis effects.

Note that hysteresis suggests a causality running from demand to supply, however, the reverse causality can also be present in the data. For instance, it is usually assumed that demand shocks ($\varepsilon_t^{\tilde{y}}$) increase GDP while increasing inflation, whereas supply shocks ($\varepsilon_t^{\bar{y}}$) increase GDP while lowering inflation. A way to help this identification of the shocks is by the inclusion of α_s, which induces a negative response on the output gap from a supply shock and by doing so induces a negative correlation with inflation via the Phillips curve (see equation (9)). In the present investigation α_s is not a trivial parameter as it is meant to capture a correlation between the trend and the output gap that can bias the estimation of η, if omitted.

Unemployment dynamics obey the following latent process.

$$u_t = \bar{u}_t + \tilde{u}_t \tag{6}$$

$$\tilde{u}_t = \beta\, \tilde{u}_{t-1} - \beta_y\, \tilde{y}_t + \varepsilon_t^{\tilde{u}} \tag{7}$$

$$\bar{u}_t = \rho_u\, \bar{u}_{t-1} + (1-\rho_u)\, u_{ss} + \eta_u\, \tilde{u}_{t-1} + \varepsilon_t^{\bar{u}} \tag{8}$$

where $\tilde{u}_t$ is cyclical unemployment and $\bar{u}_t$ is the NAIRU. The unemployment gap follows an Okun's law (see equation (7)) with β_y being the slope. The NAIRU ($\bar{u}_t$) is modeled as stationary but close to unit-root process that can potentially be affected by the cyclical component of unemployment if η_u>0. Modelling the NAIRU as a stationary but highly persistent process is grounded in various reasons. First, the sample period from 1950-2024 includes various structural changes that can be easily taken into account with a unit-root process. Second, the literature usually models this variable as a unit-root process (see Galí (2022)) due to the existing evidence on unemployment persistence (see Ball and Onken, 2022; Blanchard and Summers, 1986). The choice of modelling it with $\rho_u = 0.95$ allows for a highly persistent variable that still has defined first and second moments, which is important for the calibration of shocks variance. This choice conditions the level

of η_u. In equation (8) a gap-induced movement of the NAIRU decays at rate ρ_u, so there is some substitution between the two parameters: a similar response of unemployment can be obtained with a high ρ_u and a low η_u or with a low ρ_u and a high η_u. The literature is consistent with this: the one study that estimates a stationary NAIRU (Calvert Jump and Stockhammer, 2023) reports higher elasticities (0.49 on average) than those assuming a unit root (0.22; see Appendix C), although with a single study the comparison is only indicative. The estimates of η_u reported below should therefore be read as conditional on $\rho_u = 0.95$. Since this value is common to all economies, it bears on the level of the elasticities and not on the differences between them.

The model also includes a Phillips curve as follows:

$$\pi_t = \lambda\, \pi_{t-1} + (1-\lambda)\, \pi_{t+1|t} + \lambda_y\, \tilde{y}_t + \varepsilon_t^{\pi} \tag{9}$$

where π_t is headline inflation, $\pi_{t+1|t}$ is the rational expectation of inflation. λ is the weight of the backward-looking component of inflation expectations and λ_y is the slope of the Phillips curve. $\varepsilon t\pi$ is a cost-push shock.

Finally, monetary policy follows a Taylor rule:

$$i_t = \varphi\, i_{t-1} + (1-\varphi)\Big[\bar{r}_t + \pi_{t+1|t} + \varphi_y\, \tilde{y}_t + \varphi_\pi\Big(\pi_t - \bar{\pi}_t\Big)\Big] + \varepsilon_t^{i} \tag{10}$$

where i_t is the short-term interest rate, φ is the persistence of the Taylor rule, φ_y is the weight of the output gap and φ_π is the weight of the inflation gap with respect to the inflation target ($\bar{\pi}_t$). $\bar{r}_t$ is the natural interest rate and $\bar{\pi}_t$ is the inflation target. These two latent variables follow autoregressive processes:

$$\bar{\pi}_t = \rho_\pi\, \bar{\pi}_{t-1} + (1-\rho_\pi)\, \pi_{ss} + \varepsilon_t^{\bar{\pi}} \tag{11}$$

$$\bar{r}_t = \rho_r\, \bar{r}_{t-1} + (1-\rho_r)\, r_{ss} + \varepsilon_t^{\bar{r}} \tag{12}$$

Two simplifications of the policy block should be noted. The rule is not constrained by the effective lower bound, and for euro-area members it is specified in national variables although policy has been set for the area as a whole since 1999. In both cases the policy residual absorbs the discrepancy, so the monetary policy shocks of those episodes should

be read with caution.

The model presented above is estimated using Bayesian methods. The estimation requires proposing prior distributions for the parameters to be estimated. The prior distributions can be selected from various sources, such as a meta-analysis of the previous literature and/or expert judgment. In the next subsection, prior distributions and calibration are discussed at length.

3.1 Prior distributions

This subsection discusses the prior distributions used in the Bayesian estimation. I follow the framework of Del Negro and Schorfheide (2008) to elicit the prior distributions of the different parameters. Del Negro and Schorfheide (2008) divide the model parameters into three: the steady-state parameters, the endogenous mechanism parameters, and the exogenous parameters, as shown in the next table.

The steady-state parameters refer to those parameters that determine the steady-state solution. In the case of the new Keynesian model, there are four parameters in this subgroup: γ_{ss}, π_{ss}, u_{ss} and r_{ss}. For each of these parameters, I calibrated these parameters to be equal to the country-specific sample mean of GDP growth, inflation, the unemployment rate and the real interest rate, respectively.

The endogenous mechanism parameters are those that, intuitively, define the shapes of the impulse response functions. The discussion in the main text focuses on the elicitation of the hysteresis elasticities η and η_u as they are the novel parameters of interest. The elicitation of the remaining endogenous mechanism parameters is discussed in Appendix B. For the moment, it suffices to say that prior distributions of the endogenous mechanism parameters come from standard sources in the literature. The IS curve is inspired by the meta-analysis performed by Havránek (2015), the Phillips curve slope borrows from Smith et al. (2025) and Fitzgerald et al. (2024), the Taylor rule parameters are inspired by Taylor (1999) and Yellen (2016) and the Okun's Law parameters follow the meta-analysis done by Porras-Arena and Martín-Román (2023).

Now I turn to the elicitation process of η and η_u, the novel parameters of the model. To elicit prior distributions there are two main sources: past estimates and economic reasoning.

From economic reasoning, the law of motion (4) can be rewritten as:

$$\bar{y}_t = \bar{y}_{t-1} + \gamma_t + \eta\,\tilde{y}_{t-1} + \varepsilon_t^{\bar{y}} = (1-\eta)\,\bar{y}_{t-1} + \eta\,y_{t-1} + \gamma_t + \varepsilon_t^{\bar{y}} \tag{13}$$

Thus, $\bar{y}_t$ can be seen as a convex combination of last period's actual output and last period's potential output, meaning that η could be interpreted as the fraction of the lagged output gap that is incorporated into potential output. From this interpretation, it follows that $0 \leq \eta \leq 1$ and η can also be interpreted as a share or weight. This representation also implies that negative values of η would mechanically push potential output away from actual output, and values above one would imply overshooting, with more than 100 percent of the gap being closed in a single period. Both cases are at odds with the intended hysteresis interpretation, so it is natural to restrict η to lie in $[0, 1]$. A similar reasoning applies for η_u.

While economic reasoning restricts the parameter to some bounds, it does not inform about the shape of the prior distribution in the parameter space. In this sense the key role of past estimates is to give a sense of the most likely location of η and η_u in said parameter space. In this spirit, Appendix C surveys the available estimates of both parameters and pools them through random-effects meta-analyses to give a sense of the central tendency in the relatively small pool of existing studies; here I summarize the main findings.

For η, the evidence is scarce: beyond the suggested range of 0 to 0.25 in DeLong and Summers (2012) and the back-of-the-envelope calibrations of Rawdanowicz et al. (2014), formal estimates are only available from Jordà et al. (2020) and Roa Rozo (2024), and pooling their four reported values yields a central estimate of 0.27 with a prediction interval from 0.19 to 0.36.

For η_u, the literature is longer-standing and has more studies. I reviewed the following studies Logeay and Tober (2006), Bechný (2019), Calvert Jump and Stockhammer (2019), Calvert Jump and Stockhammer (2022), Ball and Onken (2022), Calvert Jump and Stockhammer (2023) and Clavijo-Cortes (2023). The meta-analysis delivers a central tendency of about 0.26, albeit with high heterogeneity. Moderator analysis shows that methodological choices do affect the estimates. Studies that model the NAIRU as a stationary process report significantly higher elasticities (0.49 on average) than those imposing a unit root (0.22). Similarly, studies that use Bayesian methods tend to find smaller estimates than those that use frequentist methods.

Taken together, past estimates locate both hysteresis elasticities in the low-to-middle

portion of the unit interval. Accordingly, I adopt for both η and η_u a Beta prior distribution with mean 0.3, consistent with the pooled evidence, and standard deviation 0.2, which respects the theoretical bounds derived above while remaining diffuse enough to let the country-specific data speak. Under this prior, the intervals [0, 0.2] (low hysteresis), (0.2, 0.4] (medium hysteresis) and (0.4, 1] (high hysteresis) receive prior probabilities of roughly 37%, 33% and 30%, so all regions of the theoretical interval have reasonable coverage. Section 4.4 examines the sensitivity of the posterior estimates to alternative prior distributions.

Finally, I turn to the elicitation of the prior distribution of the exogenous parameters (the shocks' variances). I follow the approach proposed by Bergholt et al. (2024) of matching the model's theoretical moments to empirical variances. Appendix D describes the approach used and the specific application to this setting.

The results of the elicitation process are found in Table 1. It was intended for all economies considered to follow the same prior distributions, however, this was not the case. In economies like Australia, the United Kingdom and South Africa the prior distributions were different in some parameters, as the standard deviations were shrunk in order to achieve convergence of the MCMC chains. Nevertheless, the prior mean is the same in all economies. In Australia, the standard deviations of the following parameters α_r, λ, φ and $\sigma_{\bar{r}}$ were shrunk to 0.2, 0.1, 0.05 and 0.5, respectively. In the United Kingdom, the standard deviation of $\sigma_{\bar{r}}$ was shrunk to 0.5. In South Africa, the standard deviations of the following parameters $\sigma_{\tilde{u}}$, $\sigma_{\bar{u}}$ and $\sigma_{\bar{r}}$ were shrunk to 5, 0.5 and 0.5, respectively.

Table 1: Proposed prior distributions

	Prior distribution				
Parameter	Law	Mean	SD	90% CI	Justification or source
A. Steady-state parameters					
γ_{ss}	Calibrated	$\overline{\Delta y}$	–		Sample average, country-specific
u_{ss}	Calibrated	$\bar{u}$	–		Sample average, country-specific
π_{ss}	Calibrated	$\bar{\pi}$	–		Sample average, country-specific
r_{ss}	Calibrated	$\overline{i-\pi}$	–		Sample average, country-specific
B. Model endogenous propagation mechanisms					
Hysteresis related (baseline)					
η	Beta	0.3	0.2	[0.02 – 0.67]	See Appendix C
η_u	Beta	0.3	0.2	[0.02 – 0.67]	See Appendix C
IS curve					
α	Beta	0.5	0.1	[0.34 – 0.66]	Demand shock's half-life close to 1 year.
α_r	Gamma	0.5	0.3	[0.11 – 1.06]	Havránek (2015)
α_s	Beta	0.3	0.025	[0.26 – 0.34]	Meant to capture nominal rigidities (Ireland, 2004).
Dynamic Okun's law					
β	Beta	0.4	0.05	[0.32 – 0.48]	Unemployment gap shock's half-life close to 1 year.
β_y	Gamma	0.2	0.17	[0.02 – 0.53]	Porras-Arena and Martín-Román (2023).
Phillips curve					
λ	Beta	0.3	0.15	[0.08 – 0.57]	Blanchard and Galí (2007, 2010)
λ_y	Inv. Gamma	0.1	0.2	[0.04 – 0.22]	Smith et al. (2025) and Fitzgerald et al. (2024).
Taylor rule					
φ	Beta	0.7	0.1	[0.52 – 0.85]	Interest rate smoothing in annual frequency.
φ_π	Gamma	1.5	0.1	[1.33 – 1.67]	Taylor (1999) and Yellen (2016).
φ_y	Gamma	0.5	0.1	[0.35 – 0.67]	Taylor (1999) and Yellen (2016).
Latent processes					
δ	Calibrated	0.99	–		Holston et al. (2017)
ρ_π	Calibrated	0.99	–		Smets and Wouters (2005)
ρ_r	Calibrated	0.99	–		Holston et al. (2017)
ρ_u	Calibrated	0.95	–		Clavijo-Cortes (2023) and Ball and Onken (2022)
C. Model exogenous propagation mechanisms					
$\sigma_{\tilde{y}}$	Inv. Gamma	1.91	10	[0.61 – 4.37]	Mean values were calibrated in order to match the median economy second moments. See Appendix D for the procedure.
$\sigma_{\bar{y}}$	Inv. Gamma	1.42	10	[0.45 – 3.27]	
σ_γ	Inv. Gamma	0.19	1	[0.06 – 0.43]	
$\sigma_{\tilde{u}}$	Inv. Gamma	0.67	10	[0.21 – 1.52]	
$\sigma_{\bar{u}}$	Inv. Gamma	0.15	1	[0.05 – 0.32]	
σ_π	Inv. Gamma	1.96	10	[0.62 – 4.48]	
$\sigma_{\bar{\pi}}$	Inv. Gamma	0.42	1	[0.14 – 0.93]	
σ_i	Inv. Gamma	2.69	10	[0.90 – 6.07]	
$\sigma_{\bar{r}}$	Inv. Gamma	0.19	1	[0.06 – 0.43]	

Source: Own elaboration.

4 Results

4.1 Estimates

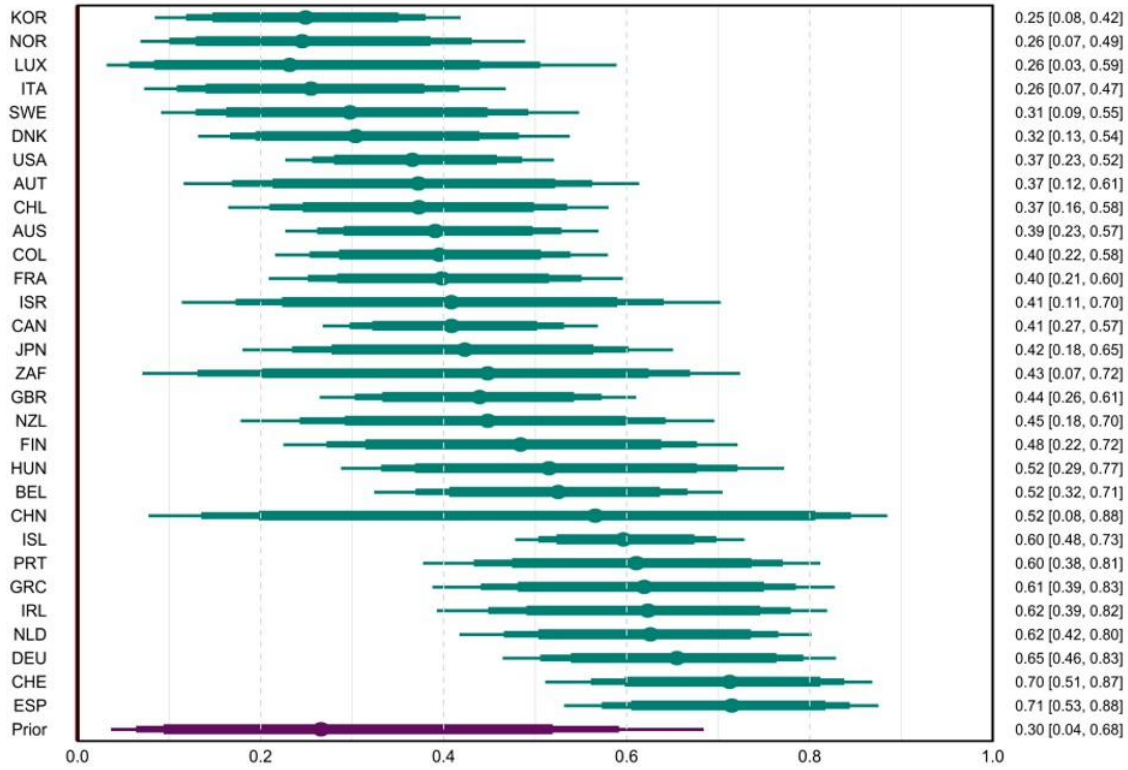
Before discussing the estimates and their implications, it is important to assess the convergence of the posterior distributions. To assess MCMC convergence, I computed Gelman and Rubin (1992) diagnostics for all structural parameters in each of the 30 country–specific Bayesian estimations. For most countries, all parameters exhibit an $\hat{R}$ below 1.05 which suggests that convergence was achieved. Results can be found in Appendix E.

I turn to some generalities about η. The cross-country average is 0.46, which is higher than the prior distribution mean and also higher than the estimates reviewed in the meta-analysis presented in Appendix C. The posterior distributions suggest an important level of dispersion across countries, which indicates that the data inform rather than simply replicating the common prior. European economies (average η of 0.50) display higher levels of hysteresis in comparison to non-European ones (average η of 0.38).

The prior distribution of η assigns about 37.6% probability to η lying in the region from 0 to 0.2. Figure 1 presents the posterior distributions of the hysteresis elasticity η for potential GDP across 30 economies studied. The posteriors provide strong evidence against small hysteresis effects in the following economies: Spain, Ireland, Iceland, Germany, the Netherlands, Switzerland, Belgium, Portugal, Canada, Greece, the United Kingdom, Hungary, the United States, Australia, New Zealand, France, Israel, Colombia, Finland, Chile, Japan, Austria, China, and Denmark. It also presents moderate evidence against small hysteresis effects for South Africa, Sweden, South Korea, Italy, and Norway where the posterior distribution still leaves non-negligible probability in the region from 0 to 0.2, but clearly less than the prior. Finally, Luxembourg is the only economy where the data are too weak or noisy to move beliefs strongly away from prior doubts about the size of hysteresis.

For η_u, the cross-country average is 0.40, which is higher than the mean of the common prior and broadly in line with, though somewhat above, the central estimates reported in the meta-analysis discussed in Appendix C. The posterior distributions exhibit substantial dispersion across economies, again indicating that the country-specific data meaningfully update beliefs rather than simply reproducing the prior. In addition, European economies display larger unemployment hysteresis on average than non-European

Figure 1: Posterior distribution of η

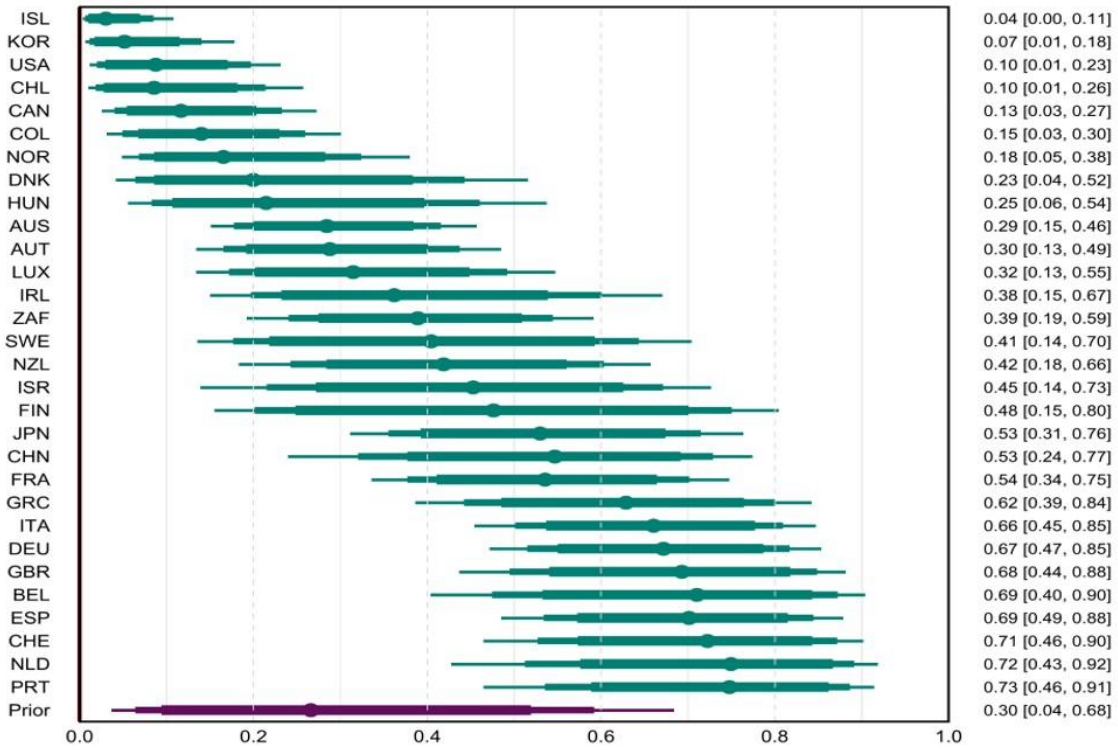


Source: Own calculations using data from Müller et al. (2025). Note: Credible intervals are derived from a 10,000 sample of the posterior distribution. The graph displays the median with a dot, the 68% credible interval with a thick line and the 90% credible interval with a thin line. The right-hand side table shows the median and the 90% credible interval in square brackets.

ones (0.48 against 0.24), suggesting that labour-market institutions and past shocks have left a more persistent imprint on the NAIRU in that group.

The prior for η_u assigns 37.6% probability to values below 0.2, but the posterior shares of said region vary widely across countries. In Iceland, South Korea, the United States, Chile and Colombia, more than 85% of the posterior mass lies below 0.2, strong evidence in favor of the independence assumption for these economies. In Canada, Norway and Denmark this share still exceeds 45%, implying that small or moderate unemployment hysteresis remains quite plausible in these economies. By contrast, in Austria, Australia, Sweden, Luxembourg and Finland the share falls to around 20–25%, and in Israel and China it drops below 0.2, while in New Zealand, South Africa, Ireland and Japan it becomes very small; in the remaining European economies—such as the Netherlands, Belgium, France, Portugal, Switzerland, the United Kingdom, Greece, Italy, Germany and Spain—the posterior mass below 0.2 is essentially zero, providing strong evidence against small hysteresis in their NAIRU dynamics.

Figure 2: Posterior distribution of η_u



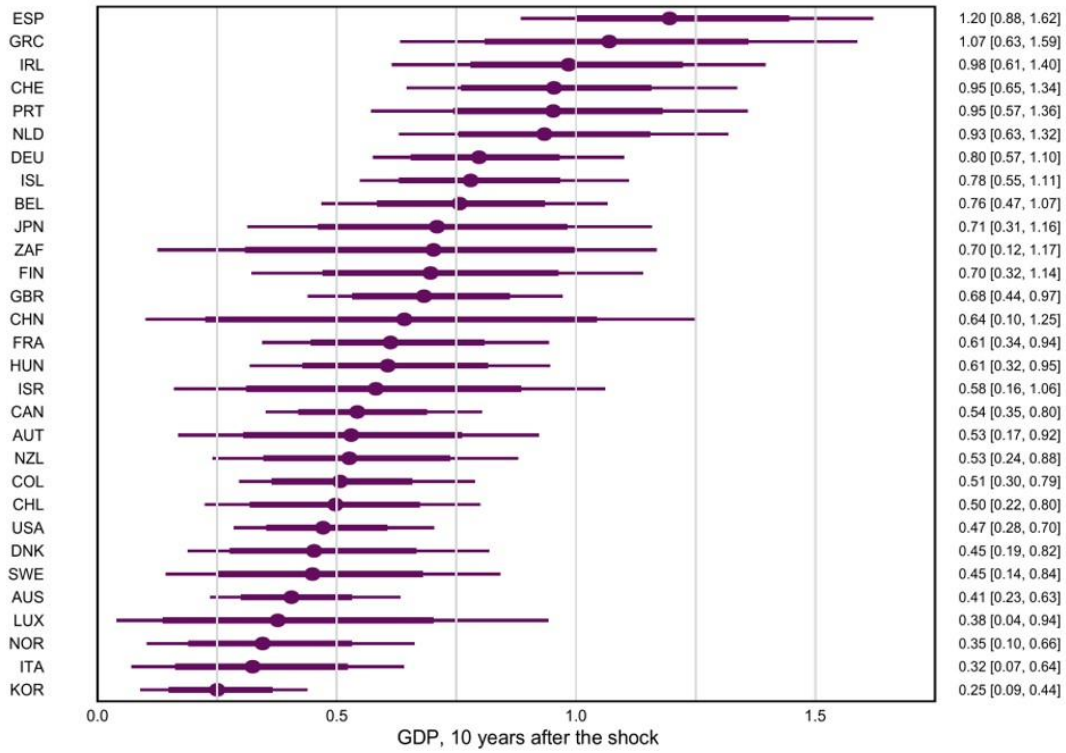
Source: Own calculations using data from Müller et al. (2025). Note: Credible intervals are derived from a 10,000 sample of the posterior distribution. The graph displays the median with a dot, the 68% credible interval with a thick line and the 90% credible interval with a thin line. The right-hand side table shows the median and the 90% credible interval in square brackets.

4.2 The long-run effects of demand and monetary policy shocks

This subsection explores the long run effect of demand shocks and monetary policy on GDP. The long run is understood as effects on the tenth year horizon. Figure 3 shows the impact of demand shocks on GDP ten years after the shock. Demand shocks were normalized to induce a one-percentage-point increase in the output gap and the GDP growth rate in the first year, making them comparable between economies. On average, the long-run effect on GDP of said demand shocks is 0.63, which is sizeable. There is substantial heterogeneity across economies. Economies like Korea, Italy and Luxembourg seem to display relatively small levels of hysteresis as the effects are on average below 0.4, while economies like Spain, Switzerland and Greece have substantial hysteresis effects as the long run effect is close to unity.

Figure 4 shows the impact of monetary policy shocks on GDP ten years after the shock. Monetary policy shocks were normalized to induce a one-percentage-point increase in the nominal interest rate in the first year, as in Figure 3, making them comparable between

Figure 3: Impact of a 1-unit positive demand shock on GDP at the 10th year



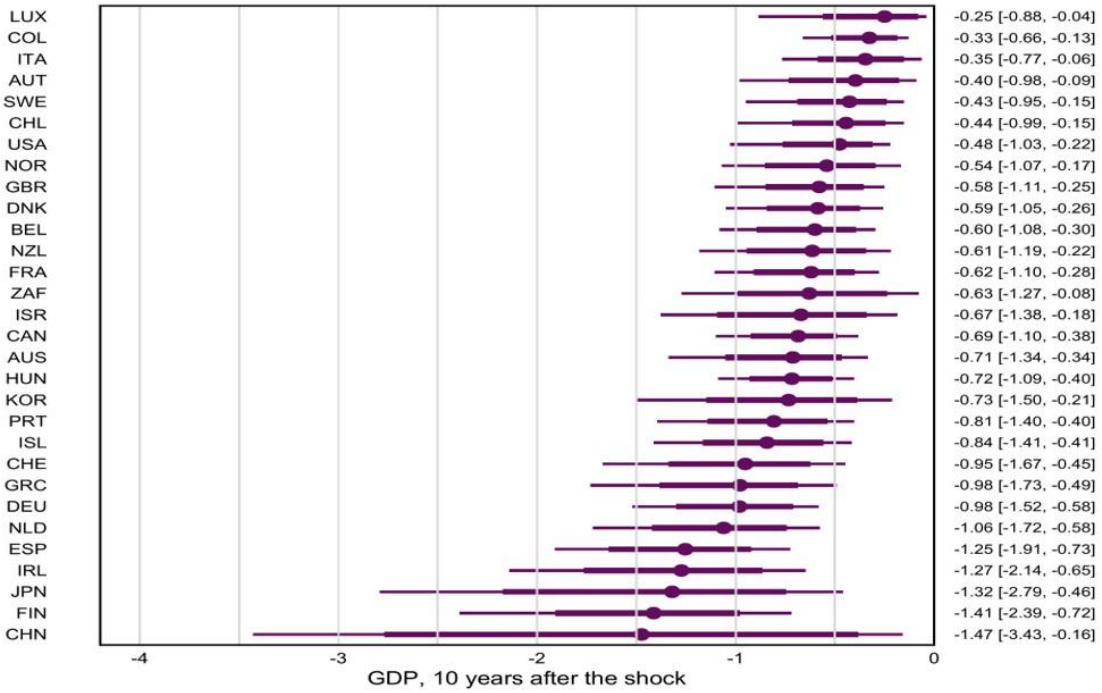
Source: Own calculations using data from Müller et al. (2025). Note: Demand shocks were normalized to induce a 1-unit increase in the output gap and the GDP growth rate in the first year. Credible intervals are derived from a 10,000 sample of the posterior distribution. The graph displays the median with a dot, the 68% credible interval with a thick line and the 90% credible interval with a thin line. The right-hand side table shows the median and the 90% credible interval in square brackets.

economies. The figure plots the posterior median of the GDP response with its 68% and 90% credible intervals for each economy. Pooling the country estimates by random-effects meta-analysis, the long-run drop in GDP from this shock is 0.635 per cent, with a 95% confidence interval from 0.538 to 0.733 that excludes zero. There is substantial heterogeneity across economies. Economies like Colombia, Italy and Luxembourg seem to display relatively small levels of monetary policy hysteresis as the effects are on average below 0.4, while economies like Japan, Ireland and China have substantial hysteresis effects stemming from monetary policy as the long-run drop exceeds unity.

4.3 Potential output and the NAIRU under hysteresis

The findings discussed in Section 4.1 suggest that hysteresis is more pervasive than previously thought. A natural question is how different the estimates of potential output and the NAIRU under hysteresis are in comparison to those where the natural rate hypothesis is assumed to hold. To answer this question, I re-estimated the model described

Figure 4: Impact of a 1-unit monetary policy shock hike on GDP in the 10th year



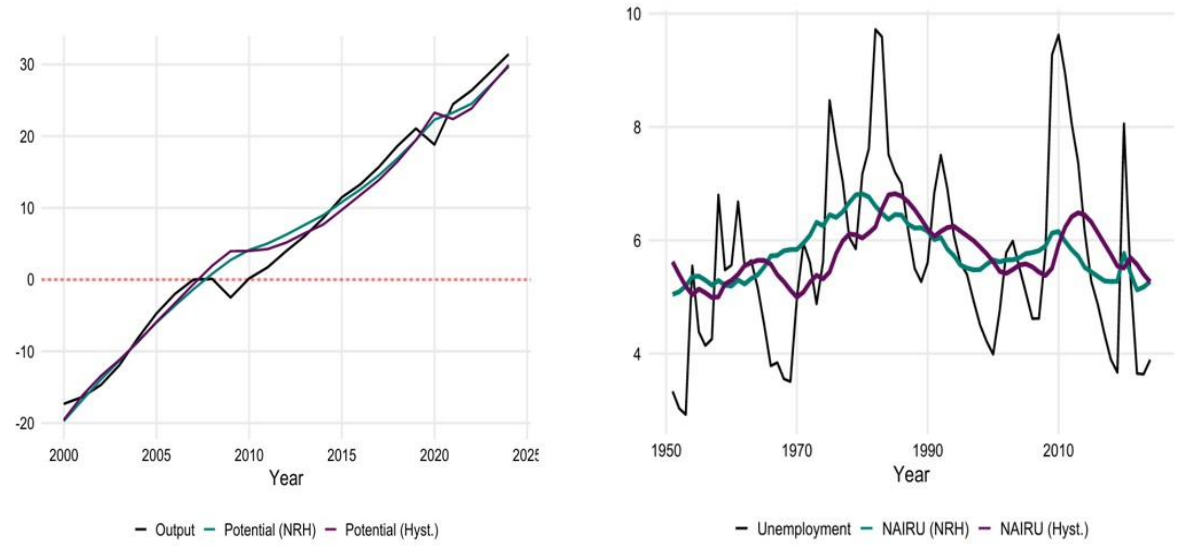
Source: Own calculations using data from Müller et al. (2025). Note: Monetary policy shocks were normalized to induce a 1-unit increase on the monetary policy rate in the first year. Credible intervals are derived from a 10,000 sample of the posterior distribution. The graph displays the median with a dot, the 68% credible interval with a thick line and the 90% credible interval with a thin line. The right-hand side table shows the median and the 90% credible interval in square brackets.

in Section 3, with the same prior distributions of Table 1 but fixing η and η_u to zero. This subsection aims to illustrate the main differences by looking into the United States data and by analysing key moments like the standard deviation and the autocorrelation of the output gap and the unemployment gap in all economies studied.

Turning to the United States, Figure 5 panel (a) depicts the evolution of GDP and potential output estimates under hysteresis and without hysteresis from 2000 up to 2024. Both measures of potential output evolve similarly from 2000 up to 2007, then during the GFC some differences arise. Firstly, potential output under hysteresis (in purple) falls faster than its counterpart with $\eta = 0$. From 2015 up to 2019 potential output under hysteresis is persistently lower than its counterpart measure. After the COVID-19 crisis, potential GDP under hysteresis is below its counterpart measure.

Figure 5 panel (b) depicts the evolution of the NAIRU estimates under hysteresis and without hysteresis from 1950 up to 2024. While from 1950 up to 1965 both measures are similar, they diverge importantly from 1965 onwards. The NAIRU under no hysteresis effects rises during 1965-1970 while the NAIRU under hysteresis falls during the same

Figure 5: Latent variables under hysteresis vs. under the natural rate hypothesis – United States



(a) Potential output estimates, 2000–2024 **(b)** NAIRU estimates, 1950–2024

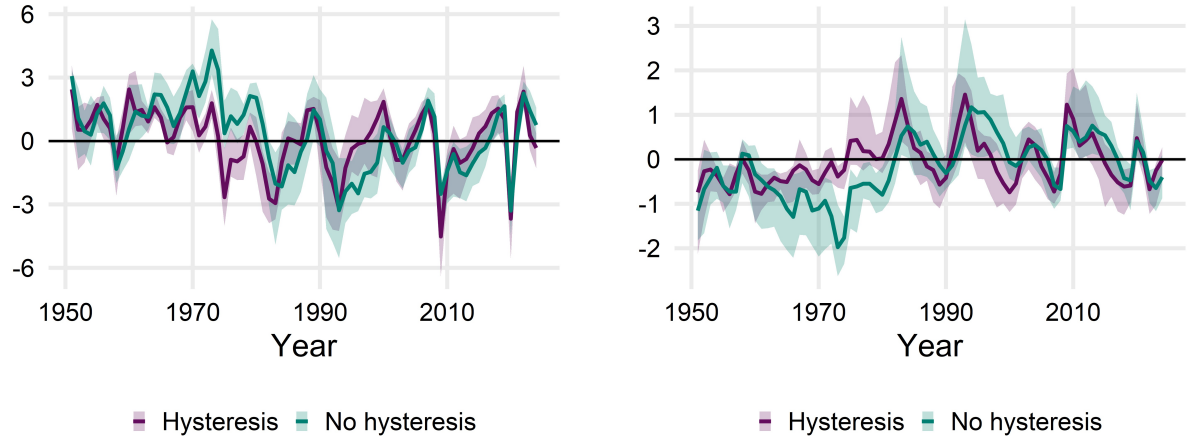
Source: Own calculations using data from Müller et al. (2025).

period. Another relevant difference is found in the aftermath of the GFC. When assuming no hysteresis effects, the NAIRU rises from 5.8 up to 6.2 in about 2 years and then decreases slowly up to 2019 around 5 percent. In contrast, when allowing for hysteresis effects, the NAIRU rises for about four years from 5.3 during 2008 up to 6.4 in 2012 and only then the NAIRU starts to fall.

Figure 6 panel (a) compares the evolution of the interquartile range and the median of the output gap for all economies studied under two scenarios: estimates under hysteresis in purple and without hysteresis in teal. In both scenarios, there seems to be an agreement in the timing of crisis, however differences are evident in their aftermath. For instance, during the crisis of 1982, 1990 and 2008 both filters find a sharp fall in the output gap, however in the aftermath the output gap closes quicker when hysteresis in potential output is allowed. The rationale is that the initial drop in the output gap is then passed into the supply side of the economy, affecting negatively its productive capacity in the next periods. A similar result is found for the unemployment gap. A gap that closes sooner could also partially explain why disinflation after the GFC was milder than the depth of the recession would have suggested.

Figure 7 compares the volatility and the persistence of the cyclical components under hysteresis versus no hysteresis. Panel a) and panel b) show the comparison of the standard

Figure 6: Output gap and unemployment gap across time



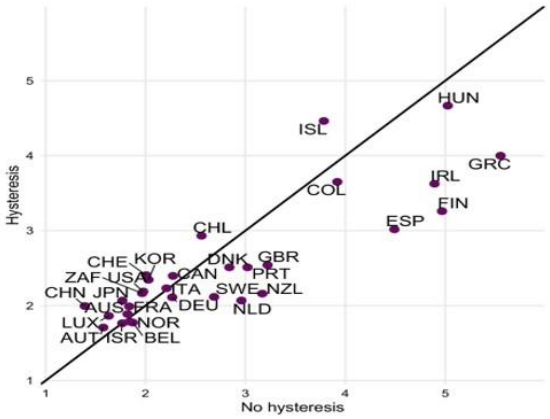
(a) Output gap **(b)** Unemployment gap

Source: Own calculations using data from Müller et al. (2025).

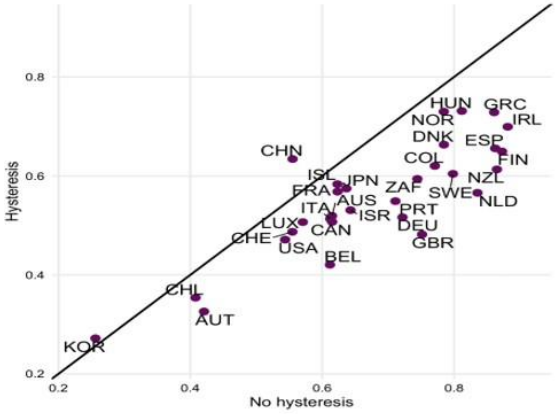
deviation and the autocorrelation of the output gap. In most countries, estimates of the output gap seem to be less volatile and less persistent under hysteresis. A similar result is found for the unemployment gap (see panel c and d). The latter finding also implies that the trend component is more volatile when allowing for hysteresis effects. By contrast, the estimates of the natural rate of interest change little between the two specifications. Unlike in Holston et al. (2017), $\bar{r}_t$ does not depend on trend growth here, so η and η_u reach it only through the filtering of the other latent variables.

A similar pattern is found when comparing the output gaps generated here with those from the IMF and the OECD. Figure 8 sets the output gaps estimated here against those published by the OECD and the IMF. Away from the crisis the three agree closely: over 1995 to 2007 the median gap averages 0.43 here against 0.48 for the OECD, and 0.55 against 0.36 for the IMF. They part company at the turning point. In 2007 I estimate a median gap of 1.6 per cent against 2.7 for the OECD and 3.3 for the IMF; in 2009 the ordering reverses, at -4.7 against -2.9 and -2.1. The official series are close to symmetric about the crisis while mine is not: the ratio of the 2007 peak to the 2009 trough is 0.94 for the OECD and 1.56 for the IMF, against 0.35 here. This is what one would expect when potential output is treated as a smooth trend that is independent of the cycle. The cycle is mean zero by construction, so a permanent decline in output that the trend is not free to absorb must be recorded as output above trend before it and below trend after; part of the pre-crisis boom is then the arithmetic residue of a level shift rather than an expansion. Under $\eta = 0$ potential can still fall, as it is a unit-root process,

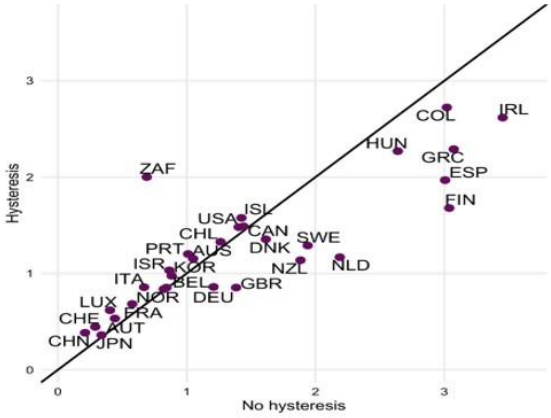
Figure 7: Comparison of volatility and autocorrelation of the output gap under hysteresis vs. the natural rate hypothesis



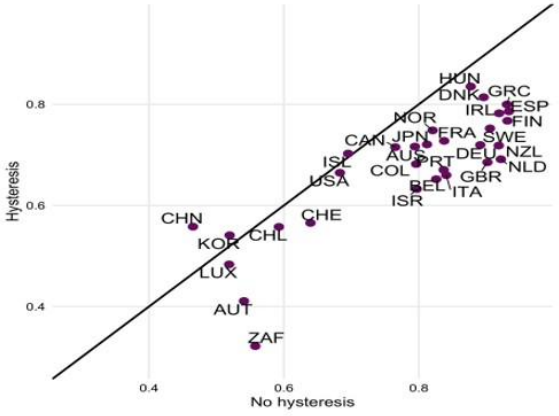
(a) Standard deviation – Output gap



(b) Autocorrelation – Output gap



(c) Standard deviation – Unemployment gap



(d) Autocorrelation – Unemployment gap

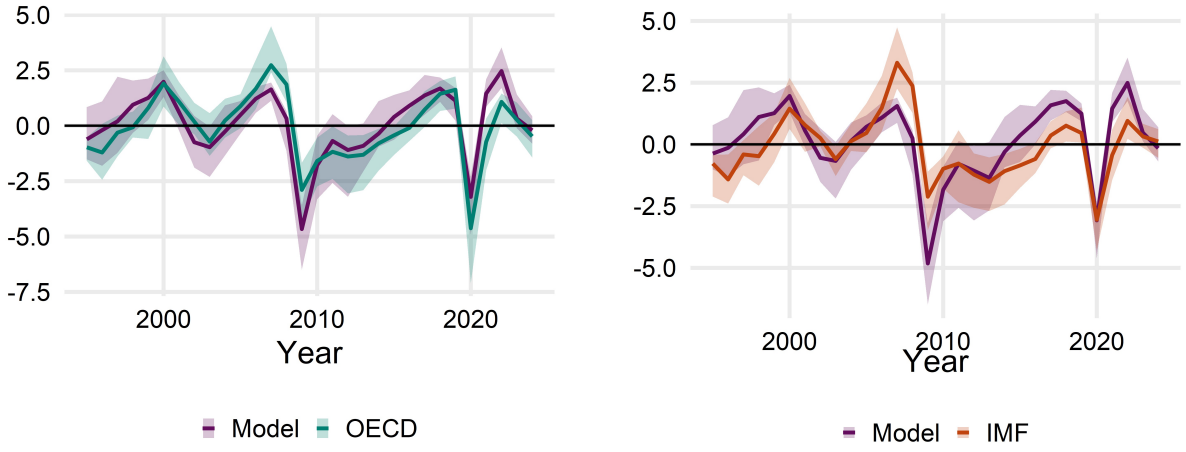
Source: Own calculations using data from Müller et al. (2025).

but only by coincidence, through a supply shock arriving in 2008. Allowing $\eta > 0$ lets potential output follow the cycle with a lag, so the decline in the trend coincides with the crisis rather than requiring an artificial boom before it. The pattern recurs, with the sign reversed, after 2020: both official series still record slack in 2021, when median inflation had risen from 1.7 to 3.1 per cent and was heading to 7.0, whereas the estimates here place the median economy above potential from 2021 onwards.

Importantly, the size of η accounts for the differences between the output gaps estimated here and those from the IMF and the OECD. In countries where hysteresis is small—Korea, Luxembourg and Italy—the estimates here are close to the official ones, correlating at 0.86 to 0.92 with the OECD over 1995–2024 and agreeing on the timing and depth of the 2008 contraction. Where hysteresis is high—Spain, Greece and Portugal—

the correlation falls to between 0.27 and 0.44: the official series record large pre-crisis booms that are absent here (4.7 and 7.7 per cent above potential in Spain and Greece in 2007, against −0.1 and 1.6), and a decade of slack that closes four to eight years earlier under hysteresis.

Figure 8: Output gaps under hysteresis vs. OECD's and IMF's output gap estimates



(a) OECD **(b)** IMF

Source: Own calculations using data from Müller et al. (2025).

The finding of a more volatile potential output and a less persistent output gap echoes the role of the correlation between trend and cycle shocks in the univariate trend-cycle decomposition literature. Morley et al. (2003) study why the trend-cycle decomposition of the unobserved components (UC) model of Watson (1986) differs so much from the Beveridge-Nelson filter, even though both aim to estimate the same object. They find that the smooth trend obtained by Watson (1986) is the result of imposing a zero correlation between the trend and cycle shocks. Once that correlation is allowed, the two filters coincide and imply a very large and negative correlation between trend and cycle shocks. This trend-cycle decomposition, however, gives rise to unintuitive output gap estimates, meaning that the highly negative correlation is difficult to defend from an economic point of view.

Subsequent work has sought to reconcile and reinterpret this result. Grant and Chan (2017) show that the discrepancy between the correlated UC model and the large, persistent cycles of the Hodrick-Prescott filter stems from how the trend is modeled: once the trend follows a second-order Markov process, intuitive output gaps are recovered at a comparable model fit. More importantly for this paper, Li and Mendieta-Muñoz (2024) argue that a non-zero correlation between trend and cycle shocks confounds two eco-

nomically distinct mechanisms of opposite direction and timing: a time-to-build effect, whereby a positive permanent shock leaves output temporarily below its new trend—generating a negative contemporaneous correlation—and hysteresis, a lagged feedback from past cycles to the trend.

The model proposed here follows the logic of Li and Mendieta-Muñoz (2024) of separating the two channels by direction and timing, but within a multivariate setting. The time-to-build effect is captured by α_s, which allows trend (supply) shocks to move the output gap contemporaneously, while hysteresis is parameterised separately by η and η_u, which map lagged cyclical conditions into potential output and the NAIRU. In addition, the Phillips curve and Okun's law provide identifying information that univariate representations lack (Appendix E assesses identification formally). Consistent with the two channels being distinct, Gonzalez-Astudillo et al. (2026) find that estimated scarring coefficients are essentially unchanged when contemporaneous trend-cycle correlation is additionally allowed for.

4.4 Robustness checks

In this subsection, I perform two main robustness checks. The first one is related to the prior distribution of η and η_u. This is needed for several reasons. First, they are quite novel parameters and while there is some empirical evidence about their mean, it is not enough to infer the shape of the distribution. Thus, aside from the prior distribution proposed in Table 1, I propose the following three distributions. The first one is to ignore previous knowledge aside from the fact that these two parameters should be bounded in the 0 to 1 interval, which can be done using a uniform distribution. The second one is Beta distribution with mean 0.3 and a standard deviation of 0.29. While it has the same mean as the baseline it places a higher weight to values close to zero, for instance, around 50% of the prior probability lies in the 0 to 0.2 interval. Finally, I used a Beta distribution with mean 0.15 and standard deviation of 0.125. It shows the influence on the results of using a relatively high and wide prior distribution for η and η_u.

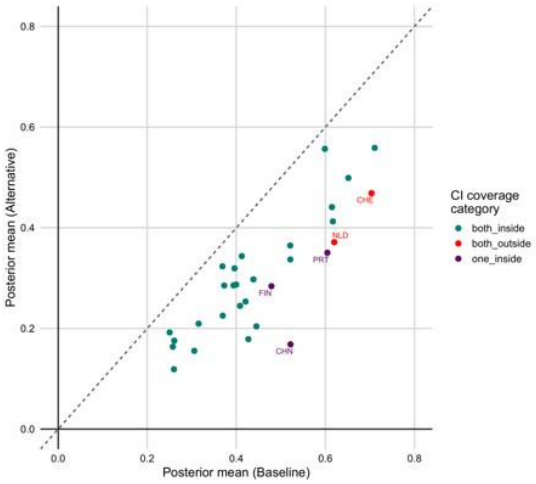
Figure 9 compares the main results of the baseline estimation with those with the alternative prior distributions of η and η_u. Posterior means under different Beta and Uniform priors preserve the ordering of countries found in the baseline, and in almost all countries the baseline posterior mean falls inside the 90% credible interval of the

alternative specification. The levels, however, are not invariant: the expected value of the prior distribution tends to influence the posterior means. In panel a) and b) most countries lie below the 45 degree line as the alternative mean is about 0.15 while in panel e) and f) the opposite happens as the prior expected value is 0.5. Despite this, the visible positive correlation in all panels suggests that countries whose data suggest high (small) hysteresis effects tend to be the same regardless of the prior assumptions. The latter implies that the data does inform the results. Figure F1 shows similar results for the long run impact of demand shock on GDP and the long run impact of unemployment gap shocks on the unemployment rate (see Appendix F).

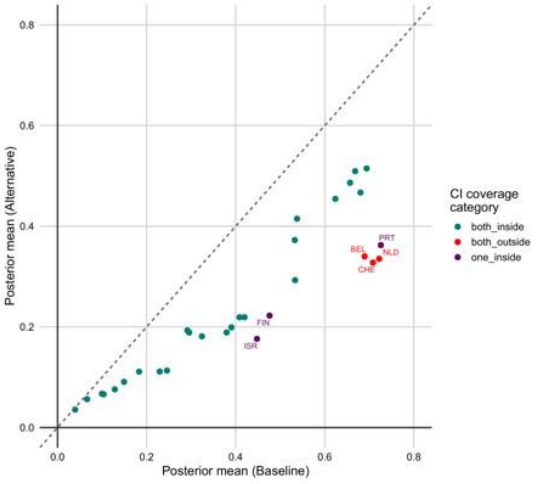
The second robustness check is related to sample period. For 17 economies that have information from 1955 or earlier, I restrict their samples to the period from 1980 onwards. This check ensures that the findings are not driven by the unique economic conditions of the 1960s and 1970s—such as the oil price shocks and the high-inflation environment—which may have operated under different structural regimes than the modern era.

Figure 10 displays the key results. Panels (a) and (b) show the posterior means for the hysteresis elasticity of potential GDP (η) and the NAIRU (η_u). Regarding η, the sample does seem to have an influence on the posterior mean, in most countries the posterior mean is lower when the sample is restricted to 1980. In the case of η_u the results are quite similar with the exception of Denmark, Israel, France and Belgium. It is important to note that there exists a clear positive correlation between the posterior mean using the whole sample and the restricted one. From this, the findings suggest that using a non-homogenous sample is not a driver of the results and that the ordering of high/low hysteresis economies is not heavily influenced by this issue. Finally, when inspecting the impact of a normalized 1-unit demand shock on GDP at the tenth year, I find relevant differences for Belgium, Denmark, France and the United Kingdom, which point to a lower degree of hysteresis in those economies under the shorter sample. Overall, these findings suggest that the ordering of economies does not depend on the sample window, although the estimated magnitudes are somewhat lower in the post-1980 sample.

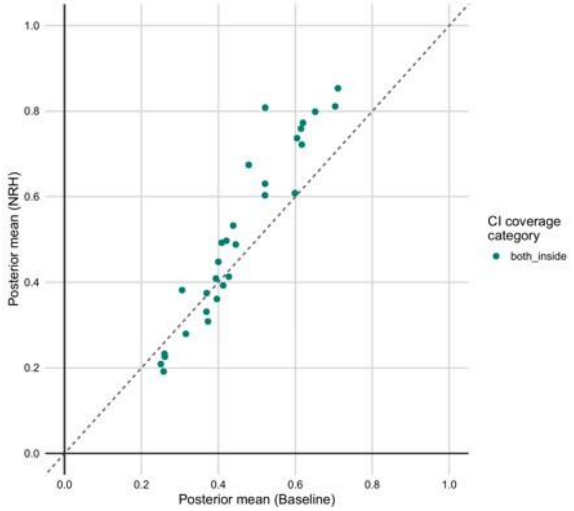
Figure 9: Robustness to changes in the prior distribution



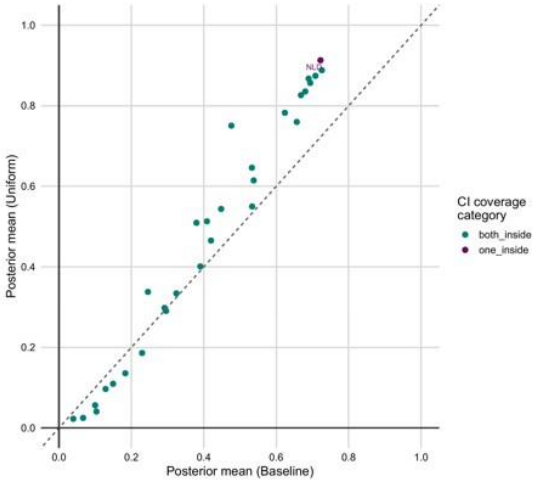
(a) η – $B(0.3, 0.2)$ vs. $B(0.15, 0.125)$



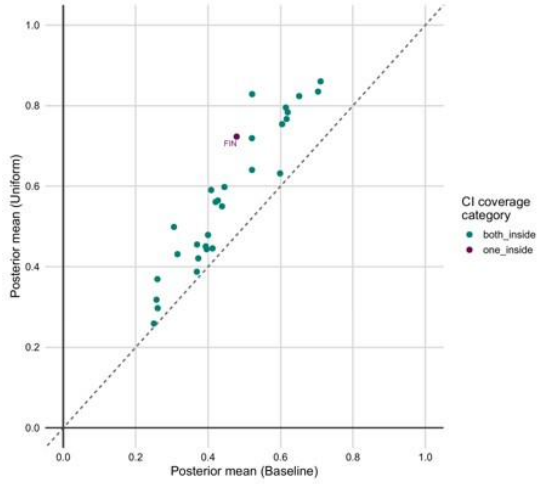
(b) η_u – $B(0.3, 0.2)$ vs. $B(0.15, 0.125)$



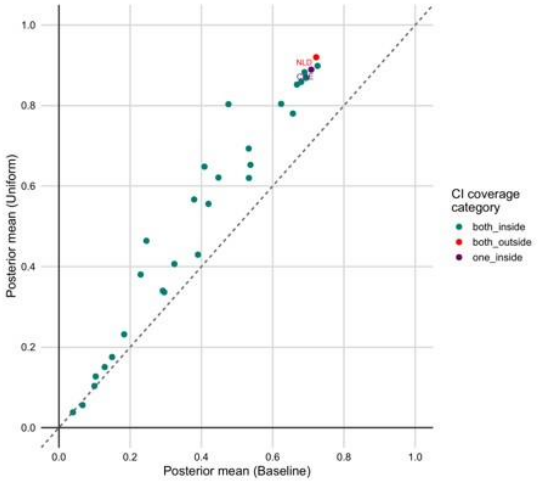
(c) η – $B(0.3, 0.2)$ vs. $B(0.3, 0.29)$



(d) η_u – $B(0.3, 0.2)$ vs. $B(0.3, 0.29)$



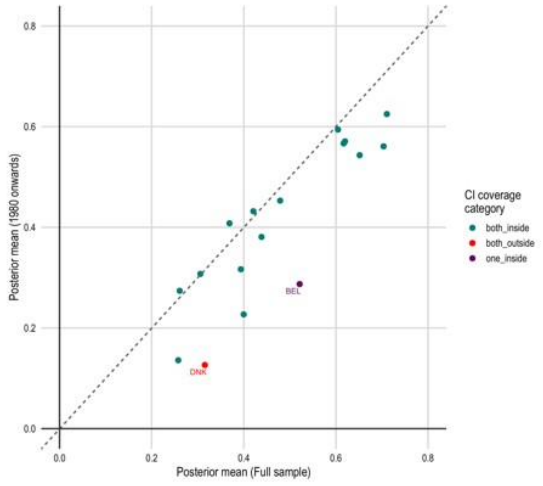
(e) η – $B(0.3, 0.2)$ vs. $U(0, 1)$



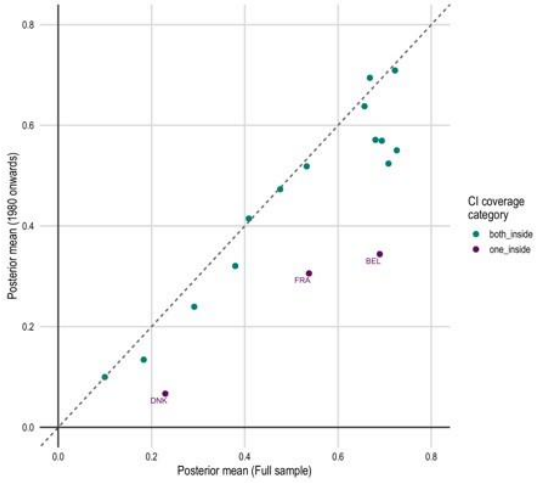
(f) η_u – $B(0.3, 0.2)$ vs. $U(0, 1)$

Source: Own calculations using data from Müller et al. (2025). Note: In all panels, teal dots represent countries where the posterior mean is included in both posterior 90% CI, purple if this only happens once, and red if both posterior 90% CI exclude the posterior mean.

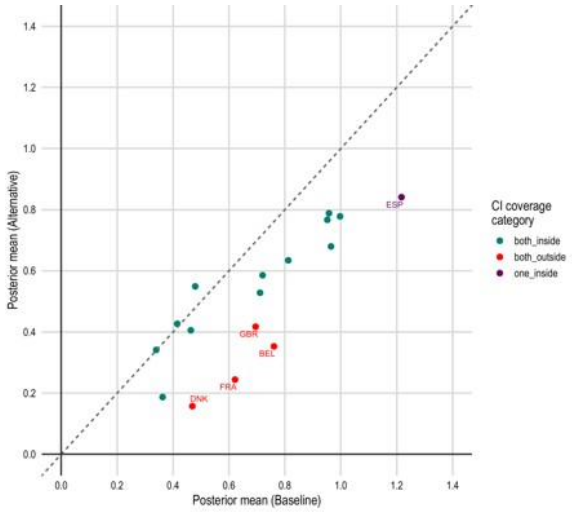
Figure 10: Robustness to changes in the sample period



(a) η



(b) η_u



(c) Demand shock IRF on GDP – posterior mean

Note: In all panels, teal dots represent countries where the posterior mean is included in both posterior 90% CI, purple if this only happens once, and red if both posterior 90% CI exclude the posterior mean.

Source: Own calculations using data from Müller et al. (2025).

5 Mechanisms

In this section, local projections are used to estimate the impact of demand and supply shocks on a number of variables outside the estimated model. This can help in understanding the mechanisms behind hysteresis effects. The use of local projections in a longitudinal setting offers some advantages. It increases statistical power and allows for the inclusion of time-fixed effects which can account for systemic shocks. Thus, local projections are suitable to answer some questions about the nature of hysteresis that the relatively simple model used here cannot.

Hysteresis in GDP was found in this study both from demand shocks and monetary policy. The question that arises is what is the source of this hysteresis. From the supply side, GDP is a function of TFP, capital, and labour. Therefore, the impact of demand shocks on these three variables is assessed to determine the main mechanisms behind it. To do so, I use the following local projection specification in a longitudinal setting.

$$X_{c,t+h} - X_{c,t-1} = \mu_{c,h} + \tau_{t,h} + \theta_h \,\text{Shock}_{c,t} + \sum_{k=1}^{3} \psi_{h,k} \,\Delta X_{c,t-k} + \epsilon_{c,t,h} \tag{14}$$

where X can be GDP, TFP, capital, CPI or nominal interest rates, where $\mu_{c,h}$ are country fixed effects, $\tau_{t,h}$ are time fixed effects, θ_h is the parameter of interest that measures the cumulated effect of a given shock on the dependent variable X in the horizon h. The regression controls for autocorrelation by including three lags of the dependent variable in first differences. In this subsection, I focus on trend shocks ($\varepsilon^{\bar{y}}$) and demand shocks ($\varepsilon^{\tilde{y}}$).

Because the shocks are estimated rather than observed, treating their posterior mean as data would understate the uncertainty around θ_h. I therefore estimate the local projection by OLS separately for each of D = 1,200 posterior draws of the shock series, in the spirit of Furlanetto et al. (2025). For every draw d and horizon h this yields a point estimate $\hat{\theta}_h^{(d)}$ and a standard error $\hat{s}_h^{(d)}$ clustered by country. Clustering matters here: the cumulative response $X_{t+h} - X_{t-1}$ makes the residual a moving average of order h by construction, and the estimated shocks are themselves serially correlated, so a variance that assumes independent errors would be too small at long horizons. Conditional on draw d, the posterior of θ_h is approximated by a Student-t distribution centred at $\hat{\theta}_h^{(d)}$, scaled by $\hat{s}_h^{(d)}$, with $G - 1$ degrees of freedom, where G is the number of coun-

tries. The posterior that integrates over the uncertainty in the shocks is the mixture $p(\theta_h \mid \text{data}) = D^{-1} \sum_{d=1}^{D} p(\theta_h \mid \varepsilon^{(d)}, \text{data})$, which I evaluate by simulation, taking 200 values of θ_h from each of the D conditional distributions and pooling them. The reported median and the 68% and 90% credible bands are percentiles of that pooled sample. Its variance is the sum of two terms, the dispersion of $\hat{\theta}_h^{(d)}$ across shock draws and the average of $(\hat{s}_h^{(d)})^2$, so the bands reflect both the uncertainty about the shocks and the sampling uncertainty of the regression coefficient.

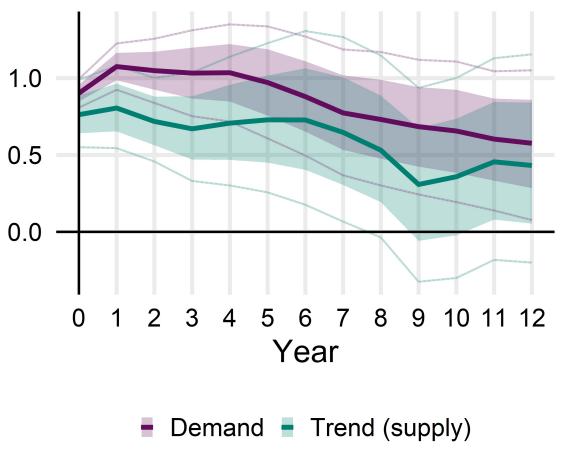
Figure 11 shows the effect of a 1-unit trend shock ($\varepsilon^{\bar{y}}$) in teal. The pattern is the one expected of a supply shock: GDP rises while prices do not. The price level is flat for most of the horizon, and its point estimate turns negative only late and imprecisely. The GDP response is itself less precisely estimated beyond the eighth year. The transmission of this shock operates mainly through a persistent increase in total factor productivity, together with an increase in the capital stock that peaks around the fifth year and fades afterwards, while employment does not respond. Nominal interest rates drift down, although this response is imprecisely estimated.

For its part, Figure 11 shows the effect of a 1-unit demand shock ($\varepsilon^{\tilde{y}}$) in purple. The pattern displayed in the figures shows the behaviour of a demand shock: both GDP and prices rise. However, the rise in GDP is persistent, contrary to the mainstream view that these types of shocks have only temporary effects. The transmission of this shock through the economy increases TFP, employment and capital, but its effect on TFP is transitory while on the other two it is persistent. In response to the increase on both GDP and prices, the policy rate rises, although the effect fades away.

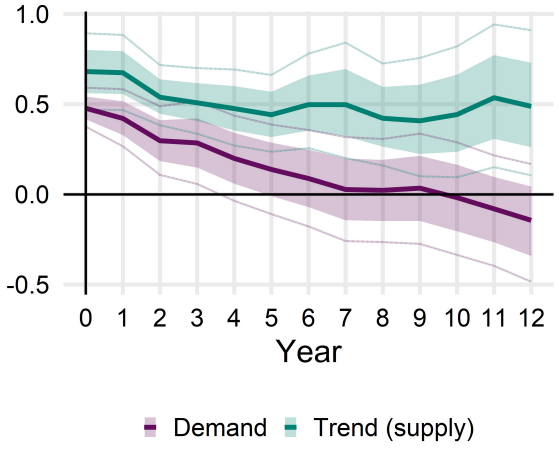
All in all, the identified supply and demand shocks do seem to behave as expected. However, as the model allows for hysteresis effects, demand shocks also have long-run effects.

A series of robustness checks were conducted, all under inference that propagates first-stage posterior uncertainty in the estimated shocks (Appendix G). Firstly, results are robust to excluding one country at a time, which guards against the pooled estimate being driven by any single economy. Secondly, results are robust to varying the lag order of the outcome controls from zero to six, which addresses the arbitrariness of the dynamic specification. Thirdly, results are robust to excluding the GFC and the COVID-19 crises. Finally, results are robust to controlling for two lags of the shock itself, which addresses

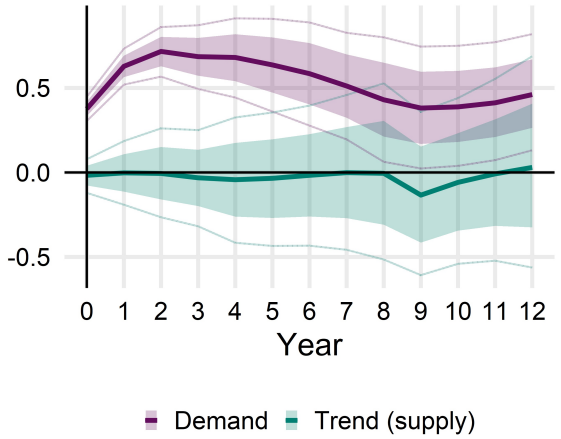
Figure 11: Effects of demand shocks and trend (supply) shocks on macroeconomic variables



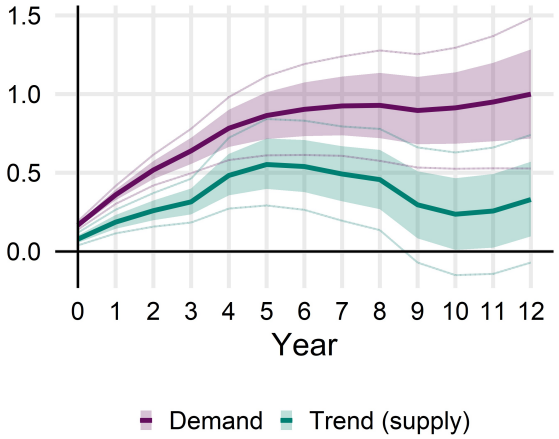
(a) GDP (in logs), percent



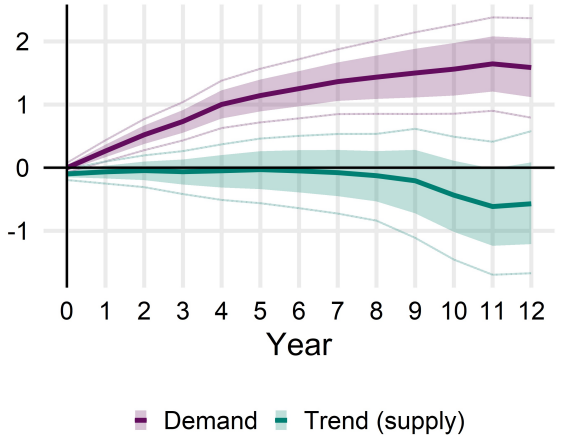
(b) TFP (in logs), percent



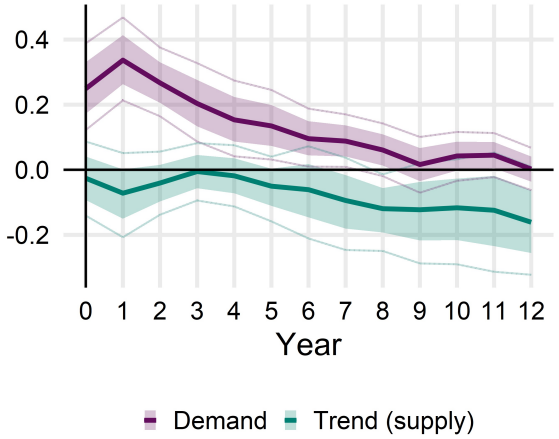
(c) Employment (in logs), percent



(d) Capital stock (in logs), percent



(e) CPI (in logs), percent



(f) Nominal interest rate, percentage points

Note: 68% credible bands are depicted as shaded areas while 90% credible bands are depicted in dotted lines.

Source: Own calculations using data from Müller et al. (2025) and the Penn World Table.

the serial correlation that smoothed shock estimates inherit by construction and could otherwise let persistence of the shock measure masquerade as persistence of the response. Across all exercises, the employment and capital responses remain persistent, with 68 percent credible bands excluding zero at horizons eight and twelve, while the long-run TFP response hovers around zero: the persistent output losses from demand shocks coincide with persistent losses in factor inputs — fewer workers and less capital — rather than in measured technology, consistent with the mechanism emphasized throughout the paper.

6 Cross-country heterogeneity

The paper has so far estimated η and η_u for a broad sample of economies, but it has said little about how they differ across economies. This section looks more closely at that heterogeneity and answers two questions. How sizeable is the heterogeneity in hysteresis? Are there structural features of the economies associated with η and η_u?

The answers to both questions draw on meta-analysis, treating each economy's posterior as one estimate of a country-specific elasticity. For the first question, meta-analysis reports statistics that give a sense of the dispersion of the true elasticities across economies, as well as a prediction of where the elasticity of an economy outside the sample would lie. For the second, meta-regressions measure how strongly the estimates are associated with structural features of the economies. Combined with the Bayesian Averaging of Classical Estimates (BACE) of Sala-i Martin et al. (2004), they help identify which variables, from a set of candidates, are most robustly associated with the parameters of interest.

One clarification is warranted before proceeding. Traditional meta-analysis aggregates estimates produced by different researchers with different methods, and treats their estimation errors as independent. The estimates here come from a single framework applied to thirty economies, which has a cost and a benefit. The cost is that a misspecification of the model would be shared by all of them, and that common shocks such as the GFC enter every economy's sample, so the estimation errors are unlikely to be fully independent. The benefit is that the same feature removes the methodological variation between studies that usually inflates heterogeneity statistics. Thus, the residual dispersion is more plausibly attributable to differences in the elasticities across economies. The interpre-

tation is not airtight, as posterior uncertainty, different sample windows and the prior adjustments of Section 3.1 remain. That is why what follows describes factors associated with the estimated elasticities and not their structural determinants.

6.1 How sizeable is the heterogeneity?

In this subsection, I use a random-effects meta-analysis to measure how large the heterogeneity in η and η_u is. The model treats each posterior mean as a noisy reading of a country-specific true value. That is, $\hat{\theta}_c = \theta_c + \varepsilon_c$. Here $\varepsilon_c \sim N(0, s_c^2)$ is the estimation error, with s_c the posterior standard deviation, and $\theta_c \sim N(\mu, \tau^2)$ is the distribution of the true elasticities across economies. Thus, the model separates the observed dispersion into two sources: estimation error, which averages out as more economies are added, and the between-country variance τ^2, which does not. The pooled mean is the precision-weighted average $\hat{\mu} = \sum_c w_c \hat{\theta}_c \big/ \sum_c w_c$, with $w_c = (s_c^2 + \tau^2)^{-1}$. To describe the heterogeneity, I rely on Cochran's Q, the I^2 statistic and the prediction interval. Cochran's Q tests whether the economies share the same elasticity, and I^2 measures the share of the observed dispersion that remains once estimation error is removed. Finally and more importantly, the prediction interval states where the elasticity of an economy outside the sample would be expected to lie. It is in the same units as the elasticity, so it can be read directly in economic terms.

Figure 12 shows the results for η, for η_u, and for their impulse-response counterparts, namely the cumulated GDP response to a demand shock at ten years and the unemployment response to a labour demand shock at five years. The pooled elasticities are 0.460 [0.408, 0.513] for η and 0.395 [0.307, 0.482] for η_u. Both are heterogeneous, but in quite different degrees. For η_u the evidence is clear: Cochran's Q rejects homogeneity ($p < 0.001$), I^2 is 81 per cent and the prediction interval runs from -0.041 to 0.831.[3] For η the statistical evidence is weaker, as Q does not reject homogeneity at the 5 per cent level ($p = 0.080$) and I^2 is 33 per cent. However, its prediction interval runs from 0.280 to 0.641, and carried to the GDP response it runs from 0.339 to 0.923. Thus, once estimation error is removed, the dispersion that remains in η is small in statistical terms

[3]Note that the synthesis is carried out on the elasticities in their natural scale and not on a logit transformation. Because the elasticities are bounded between zero and one, the normal approximation is imperfect near the bounds, which is why the prediction interval for η_u reaches slightly below zero. As the pooled means are far from either bound, the consequences are minor.

but large in economic terms.

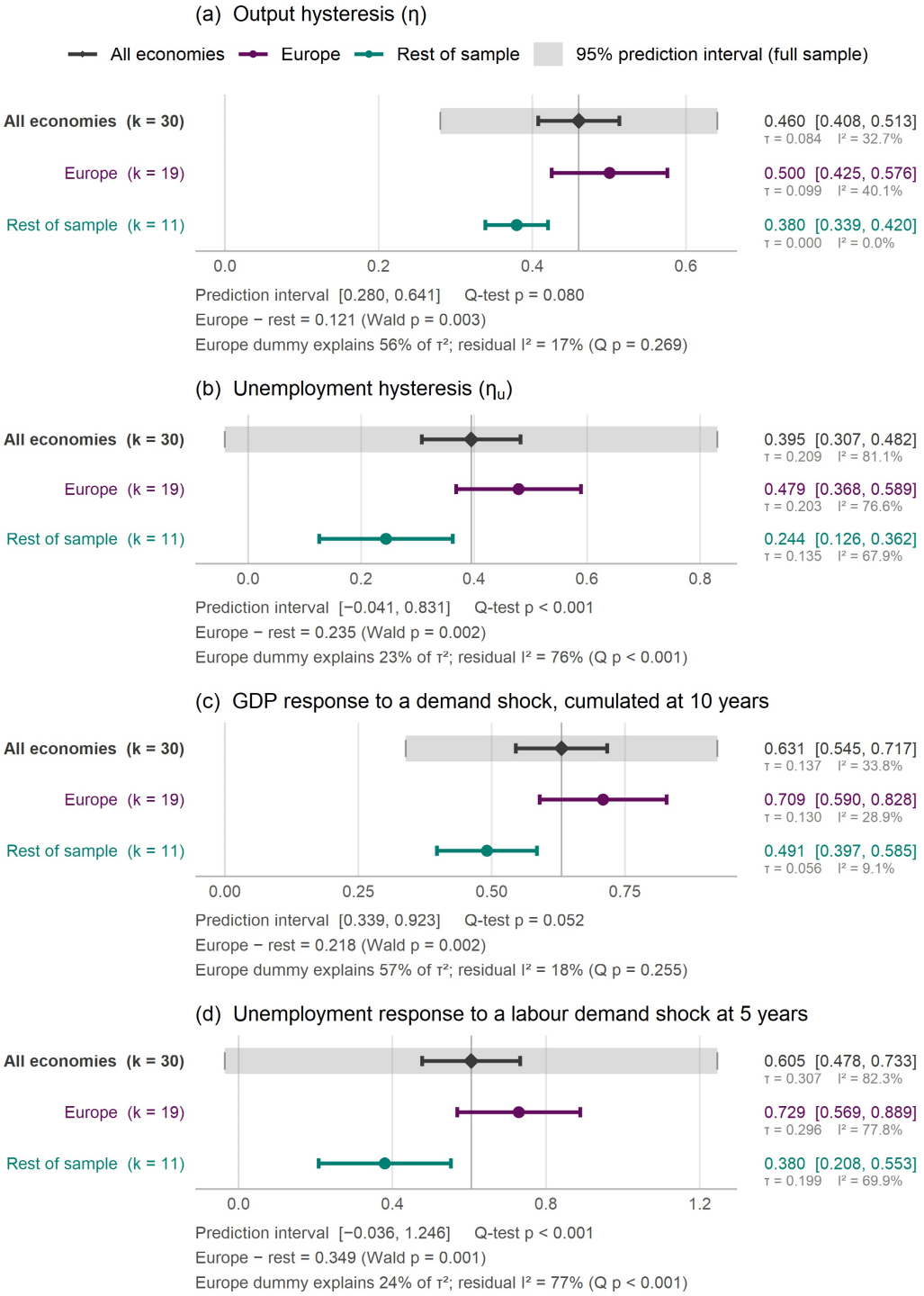
The literature has long treated hysteresis as a European phenomenon (see Blanchard and Summers (1986) and Galí (2022). Thus, comparing Europe with the rest of the sample is a natural check of whether the estimates here reproduce this stylized fact. They do. η is 0.500 in Europe against 0.380 elsewhere, and η_u is 0.479 against 0.244. The differences are 0.121 and 0.235, with Wald p-values of 0.003 and 0.002, so the regional contrast is sharper for unemployment than for output. The impulse responses preserve the contrast. Ten years after a one-unit demand shock, 0.709 of the initial GDP effect is still present in Europe against 0.491 elsewhere, a gap of 0.218 ($p = 0.002$). Five years after a labour demand shock, 0.729 of the unemployment effect remains in Europe against 0.380 elsewhere, a gap of 0.349 ($p = 0.001$). Note that the non-European GDP figure is 0.491. That is, even outside Europe, roughly half of a temporary demand shock is still in GDP after a decade. Thus, hysteresis is largest in Europe, but it is a relevant phenomenon in other regions.

The regional split shows that something distinguishes Europe from the rest of the sample, but not what. Europe differs from the other economies in key aspects: employment protection is stricter, the population is older, and monetary policy spent longer at its lower bound. All of them could potentially explain why Europe has more hysteresis than other regions. The purpose of the next subsection is to assess which explanation seems to be more relevant.

6.2 Which structural features are associated with hysteresis?

In this subsection, I use a random-effects meta-regression to relate η and η_u to the structural characteristics of the economies. The difficulty is that theory nominates more explanations than the data can adjudicate in a single regression. That is why I combine the meta-regression with Bayesian Averaging of Classical Estimates (Sala-i Martin et al., 2004). Rather than choosing one specification, BACE estimates every admissible combination of the candidate variables and weights each one by how well it fits relative to how many parameters it spends. The relevance of a variable is then summarised by the posterior probability that it belongs in the model at all, not by a coefficient conditional on one arbitrary set of controls. The obvious objection is the sample: thirty economies is not many. Note, however, that the ratio that matters is candidates to observations.

Figure 12: Random-effects meta-analysis of the hysteresis elasticities and the corresponding impulse responses, whole sample and by region



Notes: Points are random-effects pooled means and horizontal bars 95% confidence intervals; the shaded band on the top row of each panel is the 95% prediction interval for the full sample. Estimation by REML with Knapp–Hartung standard errors; the Q-test has $k-1$ degrees of freedom under the null of homogeneity. The Wald test compares the two group means without requiring them to share a τ, $z = (\mu_e - \mu_n)/\sqrt{se_e^2 + se_n^2}$, whereas the variance share and the residual I^2 come from a mixed-effects meta-regression on a Europe indicator, which does pool τ. The Q-test and the prediction interval are reported for the full sample only: a subgroup prediction interval computed on nineteen or eleven economies would be wider than the quantity's admissible range, and the within-group Q-statistics add nothing beyond the I^2 already shown. Europe is defined geographically by the following iso codes: AUT, BEL, CHE, DEU, DNK, ESP, FIN, FRA, GBR, GRC, HUN, IRL, ISL, ITA, LUX, NLD, NOR, PRT, SWE.

Source: Own elaboration based on Müller et al. (2025).

At fifteen to thirty this exercise is less demanding than the paper that introduced the method, which averaged over sixty-seven variables for eighty-eight countries, a ratio of 0.76 against 0.50 here. I cap specifications at three moderators, since a cross-section of thirty cannot estimate more saturated models reliably.

Table 2 shows the fifteen variables I enter as candidates and the mechanism behind each. What each mechanism predicts is a cross-country pattern: economies with more of the characteristic should show more hysteresis, or less, and the table records which. The groups follow six channels. A continental indicator is deliberately not among the candidates. It names no mechanism, and a regression that lets a label compete with the mechanisms it stands for answers a question about the label rather than about the mechanisms. I return to this choice below.

Formally, let $\hat{\theta}_c$ be the posterior mean of the elasticity in economy c and s_c its posterior standard deviation, and let $\mathcal{M}$ collect the models M_j formed from subsets of the $K = 15$ moderators with at most three members, so that $|\mathcal{M}| = \sum_{m=0}^{3} \binom{K}{m} = 1+15+105+455 = 576$. The space is small enough to enumerate, so every model is estimated and nothing is sampled. Model M_j is the regression

$$\hat{\theta}_c = \alpha + x'_{c,j}\beta_j + u_c, \qquad w_c = \left(s_c^2 + \hat{\tau}^2\right)^{-1}, \tag{15}$$

estimated by weighted least squares, where $x_{c,j}$ holds the moderators of M_j, each standardised to unit cross-country variance, and $\hat{\tau}^2$ is the REML between-country variance of Section 6.1. Thus, the weight is the one that produces the pooled means of Figure 12, and the two exercises treat precision alike: a loosely identified economy counts for less, but no economy counts for more than the dispersion of the true elasticities allows, which inverse-variance weights would not guarantee. This is what makes the exercise a random-effects meta-regression rather than a cross-country regression on point estimates. The constant enters every model and is not averaged over.

The prior over models is uniform across sizes and uniform across the models of a given size, $p(M_j) = \left[4\binom{K}{k_j}\right]^{-1}$ with k_j the number of moderators in M_j. It places a quarter of the prior mass on the model with no moderators, implies an expected size of $\bar{k} = 1.5$ and a prior inclusion probability of $\bar{k}/K = 0.100$ for every candidate, and penalises the multiplicity of a long candidate list automatically (Scott and Berger, 2010). Under diffuse priors on the coefficients and the error variance, the marginal likelihood of M_j relative to

Table 2: The fifteen moderators, their mechanisms and the signs theory implies

Moderator	Mechanism	η	η_u
Wage setting and insider power			
Employment protection	Firing costs make employment adjust slowly in both directions (Bentolila and Bertola, 1990). Combined with generous benefits, they lengthen unemployment spells and skills depreciate (Ljungqvist and Sargent, 1998; Blanchard and Wolfers, 2000).	+	+
Union density	Where more workers are unionised, wages are set for the employed and not for the unemployed, so equilibrium employment follows actual employment (Blanchard and Summers, 1986; Lindbeck and Snower, 1988).	+	+
Bargaining coverage	The same insider mechanism, measured by who is bound by the agreements: extension agreements also cover firms and workers who did not bargain (Layard et al., 1991; Blanchard and Wolfers, 2000).	+	+
Labour share	Proxy for insider bargaining power. Stronger insiders turn demand contractions into job losses rather than wage cuts (Blanchard and Summers, 1986; Lindbeck and Snower, 1988).	+	+
Labour reallocation and demography			
Self-employment	Self-employment absorbs displaced workers, which dampens the response of measured unemployment (Loayza and Rigolini, 2011), but it moves them into low-productivity activities from which they rarely return.	+	−
Population aged 65+	The sign is contested. A downturn may push older workers out of the labour force rather than into unemployment (Yagan, 2019). However, displaced older workers are re-employed far less often (Chan and Stevens, 2001; Neumark et al., 2019), and a shorter working horizon lowers the return to retraining, which strengthens skill depreciation (Ljungqvist and Sargent, 1998).	+	±
Capital and production structure			
Investment share of GDP	In AK-type models hysteresis works through the capital stock, so a contraction that cuts investment causes more permanent damage where investment is a larger share of output (DeLong and Summers, 1992; Reifschneider et al., 2015).	+	·
Manufacturing % of value added	Learning-by-doing concentrates productivity growth in manufacturing (Kaldor, 1966; Stadler, 1990), and its displaced workers carry specific human capital that is lost when they move to other sectors (Neal, 1995).	+	+
Innovation and product-market reallocation			
R&D % of GDP	R&D is procyclical, either because innovation concentrates where short-run profits are highest (Barlevy, 2007) or because tight credit cuts long-horizon investment first (Aghion et al., 2010). Thus, a demand contraction permanently lowers the path of technology (Stadler, 1990; Fatás, 2000; Comin and Gertler, 2006).	+	·
Product market regulation	Entry barriers sustain rents that insiders claim through wages, which raises equilibrium unemployment (Blanchard and Giavazzi, 2003). Because entry costs are sunk, a contraction that stops firm entry is reversed only slowly, and output and employment stay below trend for years (Cacciatore and Fiori, 2016).	+	+
Nominal and monetary constraint			
CPI inflation	Downward nominal wage rigidity binds more when inflation is low, as there is little inflation to erode real wages. Thus, a contraction is absorbed through employment rather than wages (Card and Hyslop, 1997).	−	−
Years at the ELB	At the lower bound, policy cannot offset the shock. Weak demand and pessimism about growth can then become self-fulfilling, and monetary policy has long-run effects (Benigno and Fornaro, 2018; Garga and Singh, 2021).	+	+
Fiscal, financial and external structure			
Government share of GDP	Automatic stabilisers reduce the amplitude of the cycle (Galí, 1994; Fatás and Mihov, 2001).	−	−
Private credit % of GDP	Where credit is scarce, downturns cut long-horizon, productivity-enhancing investment first, and financial development reduces this effect (Aghion et al., 2010). However, credit booms are followed by deeper recessions (Jordà et al., 2013).	±	±
Trade openness	Sunk costs of entering export markets make temporary shocks permanent (Baldwin and Krugman, 1989; Dixit, 1989). However, external demand gives open economies a channel for recovery.	±	±

Notes: + and − are the signs the stated mechanism implies; ± marks a moderator retained with a contested sign, where the exercise is asked to adjudicate between the two readings; · marks a channel that carries no prediction for that elasticity.

Source: Own elaboration.

the model with no moderators admits the BIC approximation

$$\log p(y \mid M_j) \;\approx\; -\frac{k_j}{2}\log N \;-\; \frac{N}{2}\log\!\left(1 - R_j^2\right). \tag{16}$$

For each moderator v, I report three statistics. The first is the posterior inclusion probability (PIP), which is the sum of the posterior probabilities of the models that contain v. The second is the inclusion Bayes factor, the ratio between the posterior odds and the prior odds of including v. A value above one means that the data increase the probability that v belongs in the model, while a value below one means that they decrease it. The third is the coefficient $\hat{\beta}$, the average of the estimates of the models that contain v, weighted by their posterior probability. Within each model, the coefficient moments are the classical weighted least squares ones, as in Sala-i Martin et al. (2004). The standard deviation of $\hat{\beta}$ includes both the sampling variance within each model and the variation of the estimates across models. Thus, it reflects uncertainty about the specification and not only about the coefficient. Both $\hat{\beta}$ and its standard deviation are expressed per cross-country standard deviation of the moderator. Finally, $P(+)$ is the share of the posterior probability of the models containing v in which the coefficient is positive. Note that this statistic says whether the sign of the association is stable, which is a different question from whether v should be included at all.

I perform two exercises to check whether these associations hold across the sample or depend on a few economies. In the first one, I repeat the BACE thirty times, leaving out one economy each time. In the second one, I exclude Chile, China, Colombia and South Africa at once, to check whether a moderator is capturing the difference between advanced and emerging economies rather than the mechanism it is meant to proxy. Tables 3 and 4 report the results for η and η_u, respectively.

Table 3 reports the results for η. As shown in Section 6.1, the dispersion of η is economically relevant, with a prediction interval from 0.28 to 0.64. However, none of the fifteen characteristics explains it. The labour share has the highest Bayes factor, 2.3, but it falls below one when Switzerland is excluded, and no other variable has a Bayes factor above one. This does not mean that the theories behind these variables are wrong. With thirty observations and a penalty that increases with the sample size, BACE is a conservative exercise and it can miss an association that exists. That is why the labour share result should be read as a weak association that future studies with more economies

could test.

Table 3: Associations between output hysteresis (η) and moderators

	Full sample ($N = 30$)			Leave one out (30 runs)			Advanced only ($N = 26$)	
Moderator	PIP	BF	$\hat{\beta}$ (s.e.)	PIP	BF	$P(+)$	PIP	BF
Labour share	0.202	2.28	+0.061 (0.027)	[0.09, 0.62]	[0.88, 14.4]	[1.00, 1.00]	0.140	1.47
Private credit % of GDP	0.096	0.96	+0.045 (0.025)	[0.05, 0.25]	[0.43, 3.06]	[1.00, 1.00]	0.070	0.68
R&D % of GDP	0.062	0.60	-0.049 (0.033)	[0.03, 0.19]	[0.29, 2.05]	[0.00, 0.02]	0.056	0.54
Self-employment %	0.033	0.30	+0.033 (0.038)	[0.02, 0.15]	[0.20, 1.58]	[0.51, 1.00]	0.032	0.29
Years at the ELB, share	0.027	0.25	+0.024 (0.029)	[0.02, 0.05]	[0.17, 0.49]	[0.88, 1.00]	0.026	0.24
Population 65+	0.027	0.25	+0.024 (0.032)	[0.02, 0.05]	[0.19, 0.46]	[0.59, 1.00]	0.024	0.22
Bargaining coverage	0.024	0.23	+0.019 (0.028)	[0.02, 0.05]	[0.19, 0.43]	[0.55, 1.00]	0.022	0.20
CPI inflation	0.024	0.22	+0.018 (0.034)	[0.02, 0.10]	[0.19, 0.96]	[0.39, 1.00]	0.026	0.24
Trade openness	0.023	0.22	+0.020 (0.029)	[0.02, 0.12]	[0.18, 1.25]	[0.45, 1.00]	0.023	0.21
Manufacturing % of value added	0.022	0.20	+0.015 (0.030)	[0.02, 0.09]	[0.16, 0.87]	[0.22, 1.00]	0.021	0.19
EPL, overall	0.021	0.20	+0.013 (0.025)	[0.02, 0.06]	[0.17, 0.56]	[0.78, 1.00]	0.021	0.20
Union density	0.020	0.18	-0.008 (0.028)	[0.02, 0.04]	[0.17, 0.39]	[0.00, 0.71]	0.021	0.19
Investment share of GDP	0.020	0.18	-0.006 (0.031)	[0.02, 0.03]	[0.17, 0.29]	[0.03, 0.97]	0.021	0.20
Product market regulation	0.019	0.18	+0.006 (0.035)	[0.02, 0.03]	[0.16, 0.28]	[0.19, 1.00]	0.020	0.18
Government share of GDP	0.018	0.17	+0.003 (0.028)	[0.02, 0.04]	[0.16, 0.35]	[0.06, 1.00]	0.019	0.17

All fifteen moderators, ordered by full-sample PIP. **Bold**: inclusion Bayes factor above 3 in every one of the 30 leave-one-country-out runs; no moderator meets this threshold for η. Grey: Bayes factor below 1 on the full sample, where the data move away from the variable.
$P(+)$ is the posterior probability of a positive coefficient given inclusion, and its range is taken over the same 30 runs. It is conditional on inclusion, so for a variable whose PIP falls below prior inclusion it is computed from models carrying negligible posterior mass: a wide range on a grey row reflects that conditioning rather than any instability in the data, and a narrow one is not evidence of a stable sign. The range is informative only for the rows in black.
The labour share leads at a Bayes factor of 2.28, but its weakest leave-one-out run, dropping Switzerland, gives 0.88.

Table 4 shows a different picture for η_u, although it should be read with the same caution. Two characteristics stand out. An increase of one standard deviation in employment protection is associated with an η_u higher by about 0.12, against a cross-country average of 0.40. Similarly, an increase of one standard deviation in the share of years at the effective lower bound is associated with an η_u higher by about 0.13. Both associations have a Bayes factor above three in each of the thirty leave-one-out runs and in the sample of advanced economies, and in both cases the coefficient is positive in practically all of the posterior probability. The share of the population over 65 has a weaker association in the full sample, but its Bayes factor rises to 7 once the four emerging economies are excluded. This suggests a demographic gradient among advanced economies that the emerging ones hide. However, as this result does not hold in the leave-one-out runs, I treat it as suggestive.

Two cautions apply. First, these are associations in a cross-section of thirty economies. The leave-one-out and advanced-only exercises check whether they depend on particular economies, but not whether they are causal. Second, the share of years at the lower bound is a coarse variable. Ten economies have a value of zero, the eleven euro-area members share the same value, and its correlation with the share of the population over 65 is 0.64. Thus, the data cannot fully separate the monetary explanation from the demographic one.

One could argue that these two variables are just a Europe indicator in disguise. The

data do not support this. Employment protection and the share of years at the lower bound each have a correlation of only 0.26 with a Europe indicator, so the indicator explains less than 7 per cent of their cross-country variance. The moderator that does follow the regional divide is the share of the population over 65 ($r = 0.71$). When the Europe indicator is added, the association of η_u with employment protection falls by 24 per cent and the association with the lower-bound share falls by 19 per cent, and both associations are at least as large within the European economies. Note, however, that the result for the lower-bound share needs care. As the eleven euro-area members share the same value, within Europe this association comes mostly from the difference between euro members and the rest, so it cannot be separated from other features of the common monetary policy.

Table 4: Associations between NAIRU hysteresis (η_u) and moderators

	Full sample ($N = 30$)			Leave one out (30 runs)			Advanced only ($N = 26$)	
Moderator	PIP	BF	$\hat{\beta}$ (s.e.)	PIP	BF	$P(+)$	PIP	BF
EPL, overall	**0.874**	**62.61**	**+0.118 (0.032)**	**[0.60, 0.97]**	**[13.30, 317.8]**	**[1.00, 1.00]**	**0.641**	**16.04**
Years at the ELB, share	**0.840**	**47.33**	**+0.134 (0.035)**	**[0.59, 0.98]**	**[12.90, 437.7]**	**[1.00, 1.00]**	**0.552**	**11.08**
Population 65+	0.154	1.63	+0.106 (0.053)	[0.04, 0.44]	[0.33, 7.11]	[1.00, 1.00]	0.448	7.31
Labour share	0.081	0.80	+0.049 (0.033)	[0.03, 0.13]	[0.23, 1.30]	[0.97, 1.00]	0.028	0.26
R&D % of GDP	0.068	0.65	-0.050 (0.037)	[0.03, 0.27]	[0.32, 3.29]	[0.00, 0.10]	0.146	1.54
Investment share of GDP	0.053	0.51	-0.039 (0.033)	[0.03, 0.14]	[0.25, 1.52]	[0.00, 0.19]	0.097	0.97
Self-employment %	0.040	0.38	+0.043 (0.047)	[0.02, 0.23]	[0.22, 2.65]	[0.96, 1.00]	0.059	0.57
Bargaining coverage	0.037	0.35	+0.035 (0.049)	[0.02, 0.12]	[0.17, 1.25]	[0.14, 0.98]	0.036	0.33
CPI inflation	0.031	0.28	-0.032 (0.054)	[0.02, 0.06]	[0.18, 0.54]	[0.00, 0.93]	0.028	0.26
Union density	0.030	0.28	-0.025 (0.036)	[0.02, 0.08]	[0.22, 0.76]	[0.00, 0.24]	0.123	1.27
Product market regulation	0.027	0.25	+0.013 (0.049)	[0.02, 0.05]	[0.16, 0.50]	[0.10, 0.99]	0.026	0.24
Manufacturing % of value added	0.025	0.23	-0.001 (0.038)	[0.02, 0.04]	[0.19, 0.35]	[0.06, 1.00]	0.026	0.24
Government share of GDP	0.024	0.22	+0.005 (0.039)	[0.02, 0.03]	[0.15, 0.27]	[0.05, 1.00]	0.022	0.20
Private credit % of GDP	0.024	0.22	+0.010 (0.034)	[0.02, 0.06]	[0.15, 0.53]	[0.06, 1.00]	0.025	0.23
Trade openness	0.023	0.21	-0.004 (0.034)	[0.02, 0.03]	[0.15, 0.28]	[0.01, 1.00]	0.021	0.20

All fifteen moderators, ordered by full-sample PIP. **Bold**: inclusion Bayes factor above 3 in every one of the 30 leave-one-country-out runs. Grey: Bayes factor below 1 on the full sample, where the data move away from the variable.
The weakest leave-one-out run excludes the United States for employment protection (Bayes factor 13.3) and South Africa for the ELB share (12.9); the strongest runs reach 318 and 438. Neither leader falls below rank 2 in any run.
Population 65+ is the only moderator that crosses a Bayes factor of 3 on the advanced subsample, rising from 1.63 to 7.31: the demographic gradient is an advanced-economy phenomenon that the four emerging economies mask.

In conclusion, the answer depends on the elasticity. For η_u the results are suggestive rather than conclusive. Economies with stricter employment protection and economies that spent longer at the effective lower bound show more unemployment hysteresis, and the share of the population over 65 adds to them only among advanced economies. The variables that are not associated with η_u are also informative. The labour share, union density and bargaining coverage, the usual proxies for insider power, all have Bayes factors below one, so the data reduce the probability that they belong in the model. Taken together, these results favour models based on firing costs and skill loss (Bentolila and Bertola, 1990; Ljungqvist and Sargent, 1998), on a constrained policy response (Benigno and Fornaro, 2018; Garga and Singh, 2021), and more weakly on the re-employment

prospects of older workers (Chan and Stevens, 2001; Neumark et al., 2019). They favour these models over the insider–outsider tradition that has organised the literature since Blanchard and Summers (1986), at least with the proxies used here. For η, none of the fifteen characteristics explains the differences across economies. Output hysteresis varies across economies by economically relevant amounts, but not along the institutional and structural characteristics that the literature has proposed. Moreover, the labour market characteristics associated with η_u are not associated with η. Note that this contrasts with Section 6.1, where the Europe indicator accounts for 56 per cent of the between-country variance of η. Thus, whatever distinguishes Europe in output hysteresis is not among the fifteen candidates considered here, and the results say which characteristics do not explain the heterogeneity of η, not that there is no explanation.

7 Conclusion

This paper provides the first country-specific estimates of the hysteresis elasticities of potential output (η) and the NAIRU (η_u) for a broad sample of 30 economies. They come from a semi-structural New Keynesian model estimated by Bayesian methods, in which both elasticities are free parameters instead of being fixed at zero. Two further exercises build on these estimates. First, the shocks identified by the model are used in panel local projections to study how demand shocks reach the supply side of the economy. Second, the country-specific posteriors of the elasticities are used in a meta-analysis and a model-averaging exercise to measure how much of the cross-country variation is real and which structural characteristics are associated with it.

The evidence favours the hysteresis hypothesis. The cross-country average of η is 0.46, so a sizeable share of the output gap is absorbed into potential output each period. Cumulated over time, this leaves 0.63 of a demand shock in GDP after ten years. The average η_u is 0.40, which implies a similarly persistent effect of cyclical unemployment on structural unemployment. The local projections indicate how this happens: employment and the capital stock respond persistently to demand shocks, while the response of measured TFP is transitory. This pattern is consistent with accumulation-based mechanisms rather than with endogenous technology.

Both elasticities vary across economies by economically relevant amounts. The 95%

prediction interval for η runs from 0.28 to 0.64, which implies a ten-year GDP effect of a demand shock between 0.34 and 0.92. As the interval excludes zero, substantial output hysteresis can be expected in an economy comparable to those in the sample, although its size cannot be predicted in advance. The interval for η_u runs from 0 to 0.83, so for an economy outside the sample a NAIRU without hysteresis cannot be ruled out. Stricter employment protection and a longer time at the effective lower bound are each associated with a higher η_u. The usual proxies for insider power are not associated with it. For η, none of the fifteen characteristics is associated with the variation.

Two implications follow. First, if demand contractions permanently reduce productive capacity, the welfare cost of recessions is larger than conventional frameworks imply, and there is a case — a conditional theoretical implication rather than a policy directive — for central banks to weigh the supply-side consequences of their stabilisation decisions (Garga and Singh, 2021). Second, the estimates are at odds with long-run monetary neutrality. Jordà et al. (2024) find that the real effects of monetary shocks last for over a decade, and show that the same persistence emerges when the responses estimated by Miranda-Agrippino and Ricco (2021), Brunnermeier et al. (2021) and Cesa-Bianchi et al. (2020) under alternative identification schemes are extended to longer horizons. This paper reaches the same result with a different method and in a wider set of economies. Under the estimated elasticities, a 100 basis point tightening lowers GDP by 0.635 per cent at the ten-year horizon, with a confidence interval from 0.538 to 0.733. That two approaches with different identification strategies and methods, and only partly overlapping samples, agree on the sign and persistence of the effect makes it less likely that either is an artefact of its own assumptions.

Several avenues for future research remain. First, allowing for "super-hysteresis" — effects on growth rates rather than levels — would show whether my estimates understate the cost of recessions. Second, the elasticities are constant over each economy's sample. If the cross-country association between employment protection and unemployment hysteresis reflects a causal link, labour market reforms would lower η_u within an economy, and the same would hold for an economy entering or leaving the lower bound. A time-varying η_u would test the cross-country evidence on a margin it cannot reach. Finally, applying the framework to economies outside the sample would test whether the prediction intervals reported here hold.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude AI in order to refine the writing of the document as English is not his first language. After using this service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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A Data sample coverage

Figure A1: Sample coverage by country



Source: Own calculations using data from Müller et al. (2025).

B Prior distribution of the remaining structural parameters

Regarding the IS curve, I assigned a Beta distribution centred at 0.5 with a standard deviation of 0.1 as the prior distribution of the output gap persistence (α). This choice reflects the view that demand shocks have short-lived effects; in this case, the half-life of a demand shock is about 1 year. In turn, the IS curve slope parameter (α_r) prior distribution is a Gamma centred around 0.5 with a standard deviation of 0.3. This distribution is inspired by the meta-analysis performed by Havránek (2015). He found that the most credible value for the elasticity of intertemporal substitution is 0.3. In

this paper, as the IS curve relates the interest rate and GDP, it is to be expected that the elasticity of intertemporal substitution will be higher as investment is more sensitive to interest rates than consumption. The proposed prior distribution aligns with this estimation. Finally, the IS curve has the parameter α_s, allowing trend shocks to negatively affect the output gap. Under sticky prices (as assumed in New Keynesian models), actual output does not adjust immediately to the new higher level of productivity; thus, while potential output rises, actual output does not rise in the same proportion. For this parameter, I used a rather informative prior distribution: a Beta distribution centred around 0.3 with a standard deviation of 0.025.

There are two key parameters in the dynamic Okun's law: the unemployment gap persistence (β) and the Okun's law slope (β_y). Regarding the persistence of the unemployment gap, its prior distribution was set to follow a Beta distribution centred at 0.4 with a standard deviation of 0.05. The latter implies that the half-life of an unemployment gap shock is around 1 year. Regarding the Okun's law slope, the prior distribution was inspired by the meta-analysis performed by Porras-Arena and Martín-Román (2023). They find that, on average, the Okun's law slope is around 0.20 when using a dynamic equation.

There are two key parameters in the hybrid Phillips curve: the weight of past inflation (λ) and the Phillips curve slope (λ_y). Regarding the weight of past inflation (λ), its prior distribution was set to follow a Beta distribution centred at 0.3 with a standard deviation of 0.15. The prior mean follows Galí and Gertler (1999) and Galí et al. (2005), who find that forward-looking behaviour dominates inflation dynamics, with a weight on lagged inflation well below one half. The standard deviation is wide enough to accommodate the higher weight, of about one half, that Blanchard and Galí (2007) estimate on annual U.S. data; see also Blanchard and Galí (2010). Regarding the Phillips curve slope, I used a back-of-the-envelope calculation based on Smith et al. (2025) and Fitzgerald et al. (2024). They find that the Phillips curve slope relating the unemployment rate and headline inflation is about 0.3. Under the dynamic Okun's law of equation (7), the long-run response of the unemployment gap to the output gap at the prior means is $\beta_y/(1-\beta) = 0.2/0.6 \approx 0.33$, so the Phillips curve slope relating the output gap and headline inflation should be around $0.3 \times 0.33 = 0.10$, which is the prior mean used.

Regarding the monetary policy rule, the interest rate smoothing parameter (φ) prior

distribution follows a Beta distribution centred around 0.7 with a standard deviation of 0.1. The prior distribution of the weight of the output gap in the Taylor rule (φ_y) is a Gamma distribution with a mean of 0.5 and a standard deviation of 0.1. The prior distribution of the weight of the inflation gap in the Taylor rule (φ_π) is a Gamma distribution with a mean of 1.5 and a standard deviation of 0.1. This choice is inspired by Taylor (1999).

Finally, the model is closed with the latent processes $\bar{\pi}_t$, $\bar{r}_t$ and γ_t, which follow AR(1) processes. The prior distributions of these parameters were chosen to reflect high levels of persistence of these latent variables. Thus, δ, ρ_π and ρ_r were set equal to 0.99, while $\rho_u = 0.95$. Regarding the latter, many contributions have assumed a unit-root process for the NAIRU, either explicitly or implicitly (e.g., Jaeger and Parkinson (1990), Logeay and Tober (2006)). More recent Bayesian studies (Bechný (2019) and Clavijo-Cortes (2023)) also adopt this assumption, often due to the long memory observed in unemployment series. I adopt the same approach, although I set ρ_u at 0.95 instead of 1. The rationale for this is that by doing so the variance of the unemployment rate and the NAIRU is defined.

C Meta-analyses for the elicitation of the prior distributions of η and η_u

This appendix documents the meta-analyses of past estimates of the hysteresis elasticities η and η_u that inform the elicitation of the prior distributions proposed in Section 3.1 of the main text.

C.1 Studies on η

The parameter η is relatively novel in the literature, and there are few estimates of it. To the best of my knowledge, η was first mentioned by DeLong and Summers (2012), who defined it as the percent reduction in the flow of future potential output per percentage-point-year of the present-period output gap and coined the term hysteresis elasticity. DeLong and Summers (2012) do not provide estimates for η, yet they suggest that η lies in the range from zero to 0.25.

Rawdanowicz et al. (2014) provided back-of-the-envelope calculations of η for OECD economies (see table 1 of Rawdanowicz et al. (2014)), and report that η could be as small as 0.08 and as high as 1.2. The average of their reported potential GDP hysteresis elasticities (η) calibrations is 0.5. Interestingly, they did not report negative hysteresis elasticities (η), even though their proposed methodology yielded negative values in some economies. This omission suggests a strong prior about the sign of this parameter.

Turning to the estimates of η, I found two sources: Jordà et al. (2020) and Roa Rozo (2024).

On the one hand, Jordà et al. (2020)[4] find that monetary policy affects TFP, capital accumulation, and the productive capacity of the economy for a very long time. To reconcile their findings with traditional economic models, they develop a theoretical model using the formulation in Stadler (1990) to characterise endogenous TFP growth in a parametrically convenient way in an extension of their small open economy New Keynesian model. The main modification to this stylised model is the inclusion of the hysteresis elasticity. Using a sample from 1870 to 2017, they estimated η=0.25 with a 95% CI from 0.21 to 0.3. On a post-war sample, η=0.67 with a 95% CI from 0.34 to 0.99.

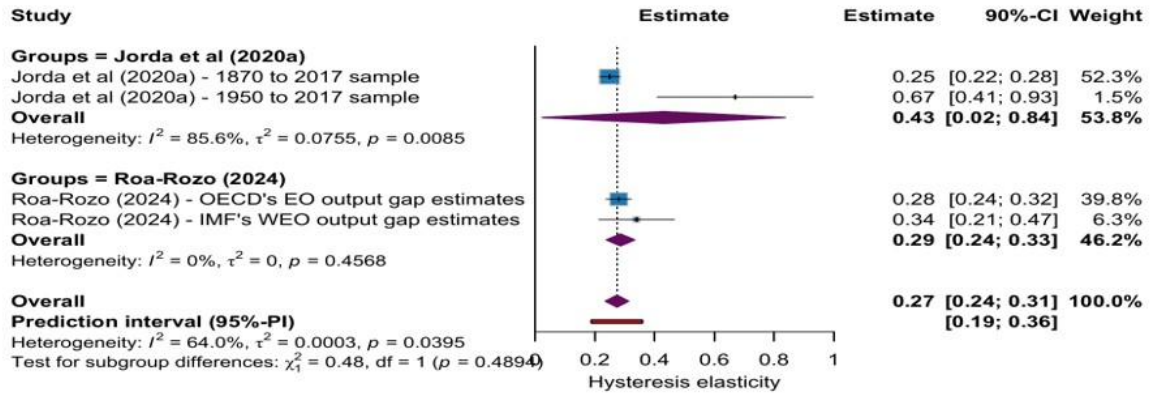
On the other hand, Roa Rozo (2024) used an error correction model to study the dynamics of GDP and potential GDP using OECD's Economic Outlook and IMF's World Economic Outlook real-time estimates of the output gap for OECD and developed economies. Using OECD estimates, he found that $\eta = 0.28$ with a standard error of 0.032, while using WEO estimates $\eta = 0.34$ with a standard error of 0.085.

A meta-analysis can help summarize the information present in literature estimates and can guide the choice of the prior distribution. All four reported estimates enter the meta-analysis — the full-sample and post-war values of Jordà et al. (2020) and the OECD and WEO values of Roa Rozo (2024) — and they are pooled with a standard random-effects model. Because two estimates come from each source, Figure C1 reports a subtotal for each source alongside the overall pooled estimate. With only two independent sources, the pooled value is best read as a summary of the available estimates rather than as a formal inference. Because Jordà et al. (2020) report 95% intervals rather than standard errors, the corresponding standard errors were recovered from those intervals (dividing

[4]It is important to note that this is not the final version of that paper (see the version published in The Review of Economics and Statistics) and that in the final version there are no estimates of η. In my view, this should not be interpreted as a sign that the estimates are not credible but rather a change in the focus of the research.

the half-width by 1.96), whereas Roa Rozo (2024) reports standard errors directly. The meta-analysis suggests that the hysteresis elasticity lies around 0.27 with a prediction interval from 0.19 to 0.36. From this exercise, the pooled value of 0.27 is rounded up to 0.30 for the prior mean, and the slight upward adjustment is immaterial given the prior SD of 0.2.

Figure C1: Meta-analysis of estimates of η



Source: Own calculations using estimates from Jordà et al. (2020) and Roa Rozo (2024).

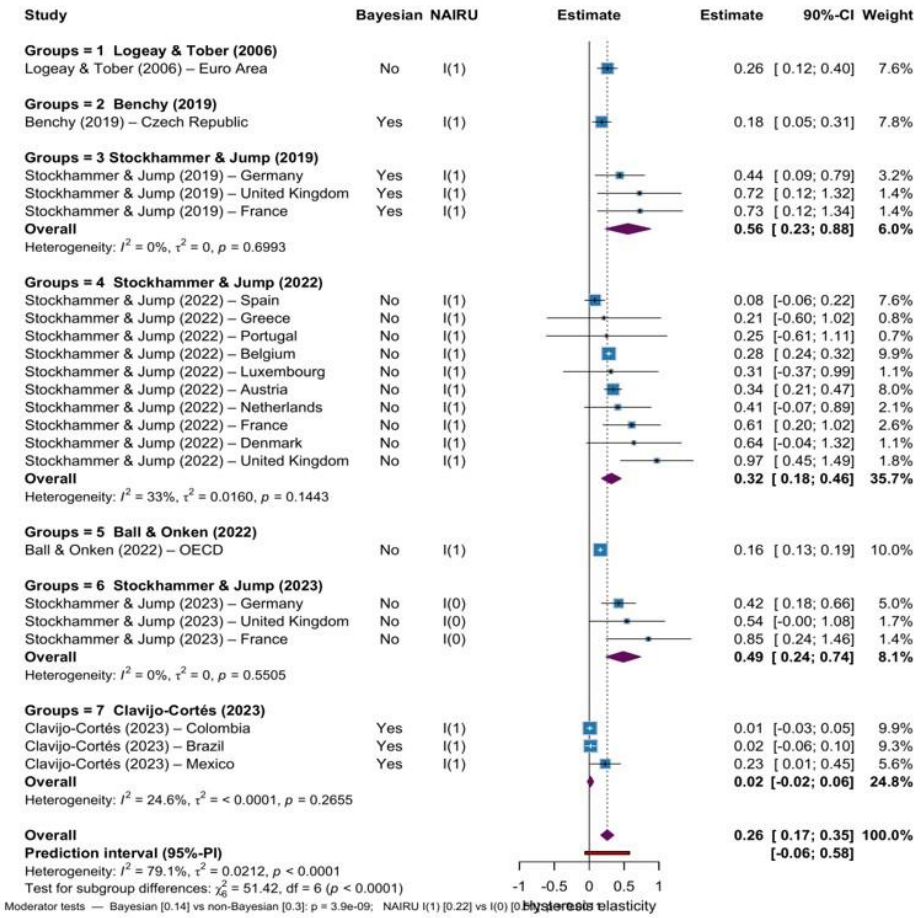
C.2 Studies on η_u

Regarding η_u, there is a longer tradition of estimating it. To my knowledge, its introduction goes back to Hargreaves Heap (1980). From there, there exist various earlier estimates of it including Jaeger and Parkinson (1990), Jaeger and Parkinson (1994) and Assarsson and Jansson (1998). After the 2000s, there are various estimates as well such as Logeay and Tober (2006), Bechný (2019), Calvert Jump and Stockhammer (2019), Calvert Jump and Stockhammer (2022), Ball and Onken (2022), Calvert Jump and Stockhammer (2023) and Clavijo-Cortes (2023). Unfortunately, pre-2000 studies cannot be included in a meta-analysis as authors do not report the standard error of the estimated η_u.

Figure C2 shows the post-2000 estimates. Some things are worth mentioning. Firstly, estimates using Bayesian methods tend to be recent (see Bechný (2019), Calvert Jump and Stockhammer (2019) and Clavijo-Cortes (2023)), being the closest to this paper. The remaining papers have used maximum likelihood estimation of unobserved component models from Jaeger and Parkinson (1990) to Calvert Jump and Stockhammer (2022).

Finally, Calvert Jump and Stockhammer (2023) used an ARIMAX approach while Ball and Onken (2022) used an error correction model in a panel setting. Bayesian estimates tend to be smaller than those that use non-Bayesian methods; one possible reading, which these few studies cannot establish, is that their prior distributions are close to the natural rate hypothesis. Another relevant difference among the studies is that in most of the literature (since Hargreaves Heap (1980)) the NAIRU is assumed to follow a unit-root process while Calvert Jump and Stockhammer (2023) do not assume it is a unit-root process and estimate its autoregressive component and find that it is highly persistent but not a unit-root process. That study reports higher hysteresis elasticities (η_u) than the rest of the literature.

Figure C2: Meta-analysis of η_u estimates



Source: Own calculations using estimates from Logeay and Tober (2006), Bechný (2019), Calvert Jump and Stockhammer (2019), Calvert Jump and Stockhammer (2022), Ball and Onken (2022), Calvert Jump and Stockhammer (2023) and Clavijo-Cortes (2023). Note: Studies that used Bayesian estimates do not report the standard deviations of the posterior distributions but do report the 90% credible interval; the standard deviation of the posterior distribution was imputed using the average distance between the median and the 90% bounds.

The meta-analysis[5] presented in Figure C2 shows a central tendency for η_u around 0.26, but with high heterogeneity (I2=79%). Estimates also differ across estimation techniques and across assumptions about the persistence of the NAIRU. The one study that does not impose a unit root on the NAIRU (Calvert Jump and Stockhammer, 2023) reports higher elasticities, 0.49 on average against 0.22 in the rest of the literature; as this contrast rests on a single study covering three economies, it cannot be separated from that study's method or sample.

[5]I excluded from the meta-analysis point estimates above 1 from Calvert Jump and Stockhammer (2019) study, as they are likely to be driven by small sample uncertainty. Including them would raise the pooled mean, so their exclusion makes the prior more conservative with respect to the hysteresis hypothesis.

D Prior distributions of the standard deviation of the exogenous variables

To derive the prior distribution of the standard deviation of the shocks, I follow the approach proposed by Bergholt et al. (2024), which selects as prior means those values that make the model replicate a set of target variances. The intuition is as follows. Because the shocks are assumed to be independent, their covariance matrix (Σ_ϵ) is diagonal, and the variance of every endogenous variable is a linear function of the nine shock variances:

$$\mathrm{Var}(\Delta y_t) = \upsilon_{11}\,\sigma^2_{\tilde{y}} + \upsilon_{12}\,\sigma^2_{\bar{y}} + \upsilon_{13}\,\sigma^2_{\gamma} + \upsilon_{14}\,\sigma^2_{\tilde{u}} + \upsilon_{15}\,\sigma^2_{\bar{u}} + \upsilon_{16}\,\sigma^2_{\pi} + \upsilon_{17}\,\sigma^2_{\bar{\pi}} + \upsilon_{18}\,\sigma^2_{i} + \upsilon_{19}\,\sigma^2_{\bar{r}}, \tag{D1}$$

$$\mathrm{Var}(\Delta \bar{y}_t) = \upsilon_{21}\,\sigma^2_{\tilde{y}} + \upsilon_{22}\,\sigma^2_{\bar{y}} + \upsilon_{23}\,\sigma^2_{\gamma} + \upsilon_{24}\,\sigma^2_{\tilde{u}} + \upsilon_{25}\,\sigma^2_{\bar{u}} + \upsilon_{26}\,\sigma^2_{\pi} + \upsilon_{27}\,\sigma^2_{\bar{\pi}} + \upsilon_{28}\,\sigma^2_{i} + \upsilon_{29}\,\sigma^2_{\bar{r}}, \tag{D2}$$

$$\mathrm{Var}(\gamma_t) = \upsilon_{31}\,\sigma^2_{\tilde{y}} + \upsilon_{32}\,\sigma^2_{\bar{y}} + \upsilon_{33}\,\sigma^2_{\gamma} + \upsilon_{34}\,\sigma^2_{\tilde{u}} + \upsilon_{35}\,\sigma^2_{\bar{u}} + \upsilon_{36}\,\sigma^2_{\pi} + \upsilon_{37}\,\sigma^2_{\bar{\pi}} + \upsilon_{38}\,\sigma^2_{i} + \upsilon_{39}\,\sigma^2_{\bar{r}}, \tag{D3}$$

$$\mathrm{Var}(u_t) = \upsilon_{41}\,\sigma^2_{\tilde{y}} + \upsilon_{42}\,\sigma^2_{\bar{y}} + \upsilon_{43}\,\sigma^2_{\gamma} + \upsilon_{44}\,\sigma^2_{\tilde{u}} + \upsilon_{45}\,\sigma^2_{\bar{u}} + \upsilon_{46}\,\sigma^2_{\pi} + \upsilon_{47}\,\sigma^2_{\bar{\pi}} + \upsilon_{48}\,\sigma^2_{i} + \upsilon_{49}\,\sigma^2_{\bar{r}}, \tag{D4}$$

$$\mathrm{Var}(\bar{u}_t) = \upsilon_{51}\,\sigma^2_{\tilde{y}} + \upsilon_{52}\,\sigma^2_{\bar{y}} + \upsilon_{53}\,\sigma^2_{\gamma} + \upsilon_{54}\,\sigma^2_{\tilde{u}} + \upsilon_{55}\,\sigma^2_{\bar{u}} + \upsilon_{56}\,\sigma^2_{\pi} + \upsilon_{57}\,\sigma^2_{\bar{\pi}} + \upsilon_{58}\,\sigma^2_{i} + \upsilon_{59}\,\sigma^2_{\bar{r}}, \tag{D5}$$

$$\mathrm{Var}(\pi_t) = \upsilon_{61}\,\sigma^2_{\tilde{y}} + \upsilon_{62}\,\sigma^2_{\bar{y}} + \upsilon_{63}\,\sigma^2_{\gamma} + \upsilon_{64}\,\sigma^2_{\tilde{u}} + \upsilon_{65}\,\sigma^2_{\bar{u}} + \upsilon_{66}\,\sigma^2_{\pi} + \upsilon_{67}\,\sigma^2_{\bar{\pi}} + \upsilon_{68}\,\sigma^2_{i} + \upsilon_{69}\,\sigma^2_{\bar{r}}, \tag{D6}$$

$$\mathrm{Var}(\bar{\pi}_t) = \upsilon_{71}\,\sigma^2_{\tilde{y}} + \upsilon_{72}\,\sigma^2_{\bar{y}} + \upsilon_{73}\,\sigma^2_{\gamma} + \upsilon_{74}\,\sigma^2_{\tilde{u}} + \upsilon_{75}\,\sigma^2_{\bar{u}} + \upsilon_{76}\,\sigma^2_{\pi} + \upsilon_{77}\,\sigma^2_{\bar{\pi}} + \upsilon_{78}\,\sigma^2_{i} + \upsilon_{79}\,\sigma^2_{\bar{r}}, \tag{D7}$$

$$\mathrm{Var}(i_t) = \upsilon_{81}\,\sigma^2_{\tilde{y}} + \upsilon_{82}\,\sigma^2_{\bar{y}} + \upsilon_{83}\,\sigma^2_{\gamma} + \upsilon_{84}\,\sigma^2_{\tilde{u}} + \upsilon_{85}\,\sigma^2_{\bar{u}} + \upsilon_{86}\,\sigma^2_{\pi} + \upsilon_{87}\,\sigma^2_{\bar{\pi}} + \upsilon_{88}\,\sigma^2_{i} + \upsilon_{89}\,\sigma^2_{\bar{r}}, \tag{D8}$$

$$\mathrm{Var}(\bar{r}_t) = \upsilon_{91}\,\sigma^2_{\tilde{y}} + \upsilon_{92}\,\sigma^2_{\bar{y}} + \upsilon_{93}\,\sigma^2_{\gamma} + \upsilon_{94}\,\sigma^2_{\tilde{u}} + \upsilon_{95}\,\sigma^2_{\bar{u}} + \upsilon_{96}\,\sigma^2_{\pi} + \upsilon_{97}\,\sigma^2_{\bar{\pi}} + \upsilon_{98}\,\sigma^2_{i} + \upsilon_{99}\,\sigma^2_{\bar{r}}. \tag{D9}$$

Each row of the system corresponds to an endogenous variable and each column to a shock, so that υ_{ij}, with $i, j = 1, \ldots, 9$, measures the contribution of one unit of variance of shock j to the variance of variable i. The υ_{ij} are functions of the structural parameters of the model only. The prior means of the shock standard deviations are then derived in

three steps: (i) choose a target value for each of the nine variances on the left-hand side; (ii) compute the υ_{ij} at the prior means of the structural parameters; and (iii) solve the resulting 9×9 linear system for the shock variances.

Step 1: target variances. For the observed variables, I use sample moments. $\text{Var}(\Delta y_t)$ is set equal to the median sample variance of GDP growth across the thirty economies studied, and the targets for the unemployment rate (u_t), inflation (π_t) and the nominal interest rate (i_t) are obtained in the same way. For the unobserved variables $\Delta\bar{y}_t$, γ_t, $\bar{u}_t$, $\bar{\pi}_t$ and $\bar{r}_t$, the targets are set using the HP filter or economic reasoning. For γ_t, I estimated the trend component of GDP growth in each economy using an HP filter ($\lambda = 100$, as the data are annual), computed the variance of that trend component, and took the median across the thirty economies as the target. I proceeded in the same way for $\bar{\pi}_t$, using the trend component of inflation. For $\Delta\bar{y}_t$, the variance should obey the inequality $\text{Var}(\Delta y_t) > \text{Var}(\Delta\bar{y}_t) > \text{Var}(\gamma_t)$, since potential growth is smoother than observed growth but more volatile than its own trend component; I therefore set $\text{Var}(\Delta\bar{y}_t)$ equal to the average of the targets for $\text{Var}(\Delta y_t)$ and $\text{Var}(\gamma_t)$. For $\bar{u}_t$, I set the target equal to 0.75 of the median sample variance of the unemployment rate, reflecting the view that most low-frequency variation in unemployment is structural. Finally, as data on inflation expectations are scarce both in the pre-sample and in the sample period, the target for the variance of the natural interest rate $\bar{r}_t$ was set equal to a tenth of the target variance of the nominal interest rate.

Step 2: computing υ_{ij}. Deriving an analytical expression for each υ_{ij} is cumbersome, but numerical values are straightforward to obtain using the one-at-a-time framework (see Saltelli et al., 2008). Because the shocks are independent, the variance of each endogenous variable is exactly linear in the shock variances, so perturbing one shock variance at a time recovers each column of υ_{ij} without approximation. As Dynare computes the theoretical second moments of the model for given parameter values, I proceed as follows, with the structural parameters fixed at their prior means:

- First, compute the variances of the endogenous variables in a baseline case where all nine shock variances are equal to one. I denote these $\text{Var}_{base}(\cdot)$.
- Second, set the variance of the demand shock to $\sigma^2_{\tilde{y}} = 2$, keep the other eight equal

to one, and compute the variances of the endogenous variables again. I denote these $\text{Var}_{\sigma^2_{\tilde{y}}}(\cdot)$.

- Third, return $\sigma^2_{\tilde{y}}$ to one, set the variance of the trend shock to $\sigma^2_{\bar{y}} = 2$, and compute $\text{Var}_{\sigma^2_{\bar{y}}}(\cdot)$.
- Repeat the process for each of the remaining shocks, in the order of the columns of equations (D1) to (D9), until $\sigma^2_{\bar{r}} = 2$.

Since each perturbation raises a single shock variance by exactly one unit, the first column is $\upsilon_{i1} = \text{Var}_{\sigma^2_{\tilde{y}}}(\cdot) - \text{Var}_{base}(\cdot)$, the second is $\upsilon_{i2} = \text{Var}_{\sigma^2_{\bar{y}}}(\cdot) - \text{Var}_{base}(\cdot)$, and so on for the remaining seven, where in each case the difference is evaluated for the nine variables i on the left-hand side of the system. This delivers all the elements υ_{ij} for any given parametrisation of the model's endogenous propagation mechanisms.

Step 3: solving for the shock variances. With the left-hand side of equations (D1) to (D9) and all υ_{ij} known, the only unknowns are the nine shock variances $\sigma^2_{\tilde{y}}$,...,$\sigma^2_{\bar{r}}$, which are found by solving the 9×9 linear system. A unique solution exists whenever the matrix of υ_{ij} is nonsingular. Since the system is solved for variances, the prior mean of each shock's standard deviation is the square root of the corresponding solution. Not every solution is admissible, as variances cannot be negative; if the system delivers a negative variance, either the prior means of the structural parameters or the target variances must be revised. In the calibration reported in Table 1, all nine variances were positive, so no revision was needed.

E Identification diagnostics

Identification is an important problem in DSGE models, which are akin to the one proposed here. Identification requires that the objective function must have a unique zero and should display "enough" curvature in all relevant dimensions. Lack of identification can lead to some relevant problems such as observational equivalence, under-identification, partial identification, and weak identification (see Canova and Sala, 2009). In practice, these problems have important consequences as they can lead to wrong conclusions from estimation, rendering inference unreliable. Moreover, in the Bayesian framework, if parameters are not identifiable the prior distribution becomes extremely influential, which can make comparisons of the prior and posterior distributions misleading. Fortunately, parameter identification is a model property and can be readily assessed on a case-by-case basis before taking the model to data.

This appendix assesses the identification of the model's parameters using the diagnostic tests readily available in DYNARE. Figure E1 shows that all the parameters in the model are locally identified at the prior mean, with no rank deficiency. However, it is important to stress that the key parameters of interest η and η_u do not have the strongest identification in comparison to other parameters. This result is reasonable as those parameters relate to the effects of an unobservable variable (output gap or unemployment gap) on another unobservable variable (potential growth or the NAIRU). However, there are two ways of improving the identification strength. The first one is to reduce the set of parameters to be estimated by omitting those with relatively high correlation with η and η_u. The second one is to introduce outside information about these unobservables.

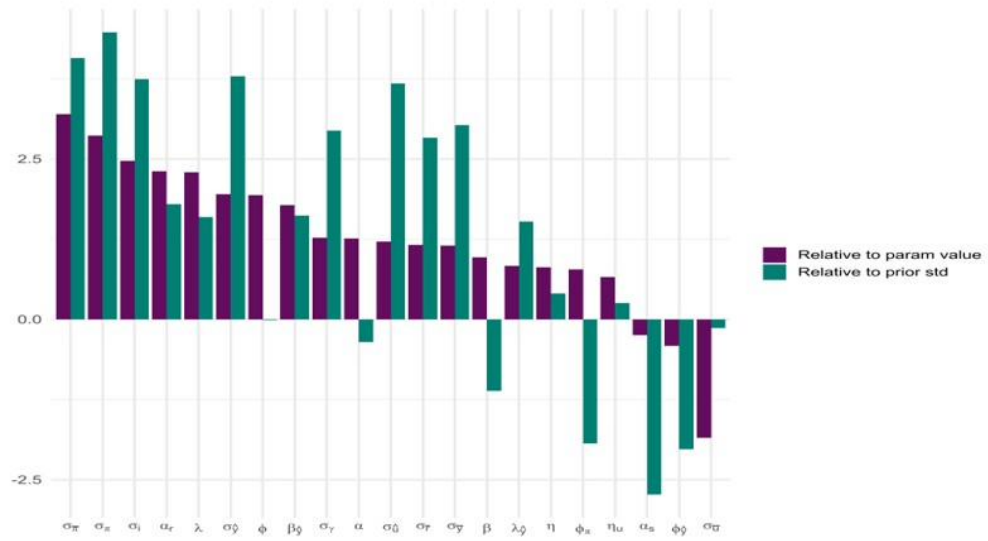
Regarding the first approach, in Section 4.4 I perform a robustness check assessing a sensitivity analysis to the prior distribution used. I believe this also serves a similar identification diagnostic purpose: if the posterior means of η and η_u were driven by the prior rather than by the data, the cross-country ordering would follow the prior and be unstable across specifications; it is not. This speaks to the informativeness of the data about the two elasticities, though not to their separate identification from the remaining parameters, which would require tracking how the posteriors of those parameters move when the prior of η and η_u changes. Regarding the second one, it is difficult to implement, while there exist estimates of the NAIRU or the potential GDP growth (see Ball and Onken (2022), Roa Rozo (2024), and Celik et al. (2023)), those estimates have short time

spans, and those estimates might be contaminated by the implicit views on hysteresis from the analysts. Therefore, I left this type of exercise for future research.

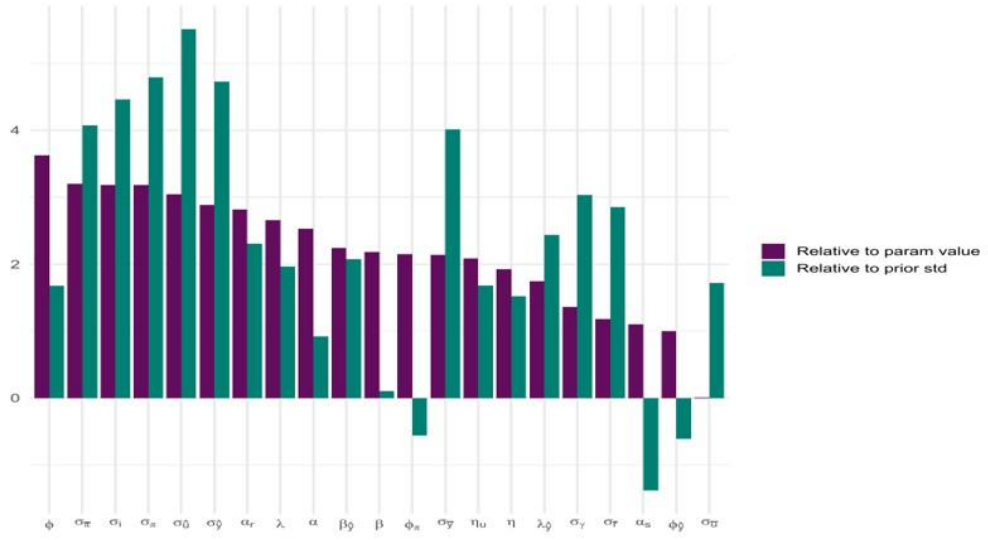
A question that must be addressed is what variation in the data helps identify η and η_u. The primary source of identification operates through the Phillips curve. A persistent decline in GDP accompanied by stable inflation is inconsistent with a large negative output gap, as the latter would generate sustained disinflationary pressure via equation (9). In such episodes, the model infers that potential GDP itself declined, attributing the stability of inflation to a smaller output gap. This mechanism is most clearly operative in economies that experienced deep recessions without commensurate disinflation, such as Spain and the United Kingdom after 2008. To a lesser extent, this identification channel can also be served by exogenous trend shocks; however, the IS curve (3) provides partial separation, as trend shocks simultaneously lower potential and raise the output gap, generating different co-movement patterns between output, inflation, and interest rates than the hysteresis channel. Two additional sources of identification reinforce this mechanism. First, η and η_u affect the persistence of GDP growth and unemployment respectively, with higher values generating excess serial correlation beyond what the IS curve persistence (α) alone can produce. Second, the model implies that η and η_u alter the cross-correlation structure of the observables — for instance, higher η tends to strengthen the negative co-movement of GDP growth and unemployment, and to increase the co-movement of GDP growth and inflation.

To complement the analytical identification diagnostics, Figures A9–A12 of the online appendix present mode check plots for η and η_u across all 30 economies. These plots display the log-likelihood kernel and the log-posterior as functions of each parameter, evaluated in a neighborhood around the estimated mode. In the vast majority of economies, both the likelihood and the posterior exhibit well-defined concavity around the mode, confirming that the data contribute curvature to the objective function.

Figure E1: Identification diagnostics



(a) Identification strength with asymptotic information matrix (log scale)



(b) Sensitivity component with asymptotic information matrix (log scale)

Source: Author's calculations with the aid of Dynare's identification toolbox.

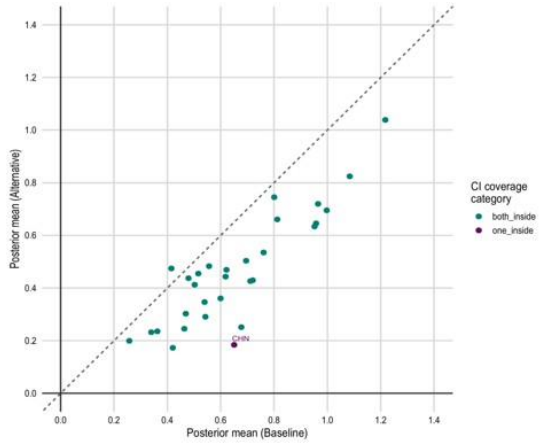
Table E1: Gelman & Rubin's $\hat{R}$ statistic

Country	$\sigma_{\tilde{y}}$	$\sigma_{\bar{y}}$	σ_{γ}	$\sigma_{\tilde{u}}$	$\sigma_{\bar{u}}$	σ_{π}	$\sigma_{\bar{\pi}}$	σ_{i}	$\sigma_{\bar{r}}$	η	η_{u}	α	α_{r}	α_{s}	β	β_{y}	λ	λ_{y}	φ	φ_{π}	φ_{y}	MPSRF
AUS	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
AUT	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
BEL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
CAN	1.03	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.09	1.00	1.00	1.00	1.02	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.00	1.01
CHE	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
CHN	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.04	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.02	1.00	1.00	1.00	1.01
COL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
DEU	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
DNK	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.00	1.00
ESP	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
FIN	1.02	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.04	1.00	1.00	1.00	1.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01
FRA	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.06	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01
GBR	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
GRC	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.09	1.00	1.00	1.00	1.04	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01
HUN	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
IRL	1.00	1.00	1.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
ITA	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
JPN	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
LUX	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.00	1.02	1.00	1.00	1.00	1.04	1.00	1.00	1.00	1.00	1.03	1.02	1.00	1.00	1.02
NLD	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.00	1.01
NOR	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
NZL	1.00	1.00	1.02	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.00	1.01
PRT	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SWE	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01
USA	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
ZAF	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
CHL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.03	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01
ISL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
ISR	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.03	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.01
KOR	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

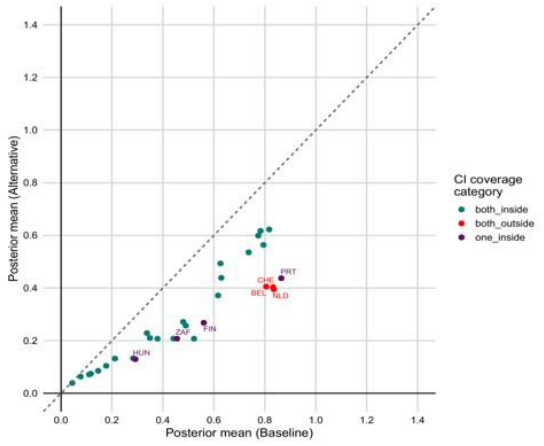
Source: Author's calculations.

F Complementary robustness checks

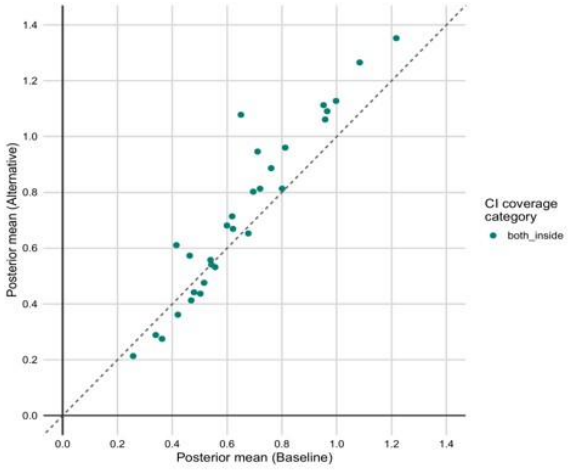
Figure F1: Robustness to changes in the prior distribution – demand shock IRFs



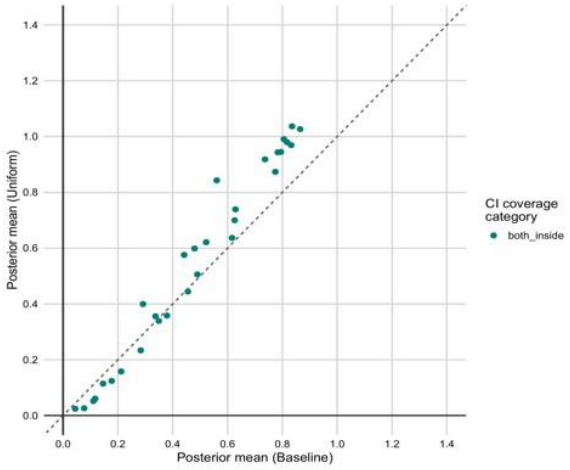
(a) η – $B(0.3, 0.2)$ vs. $B(0.15, 0.125)$



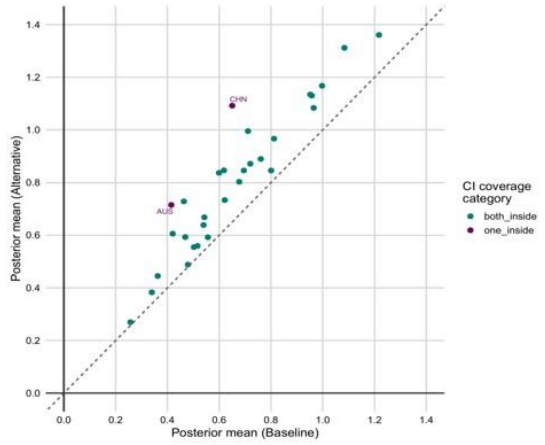
(b) η_u – $B(0.3, 0.2)$ vs. $B(0.15, 0.125)$



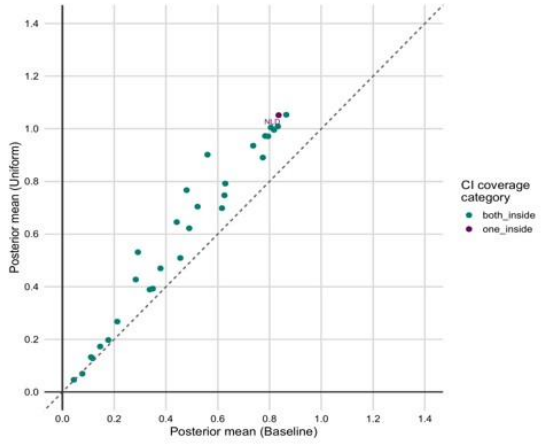
(c) η – $B(0.3, 0.2)$ vs. $B(0.3, 0.29)$



(d) η_u – $B(0.3, 0.2)$ vs. $B(0.3, 0.29)$



(e) η – $B(0.3, 0.2)$ vs. $U(0, 1)$



(f) η_u – $B(0.3, 0.2)$ vs. $U(0, 1)$

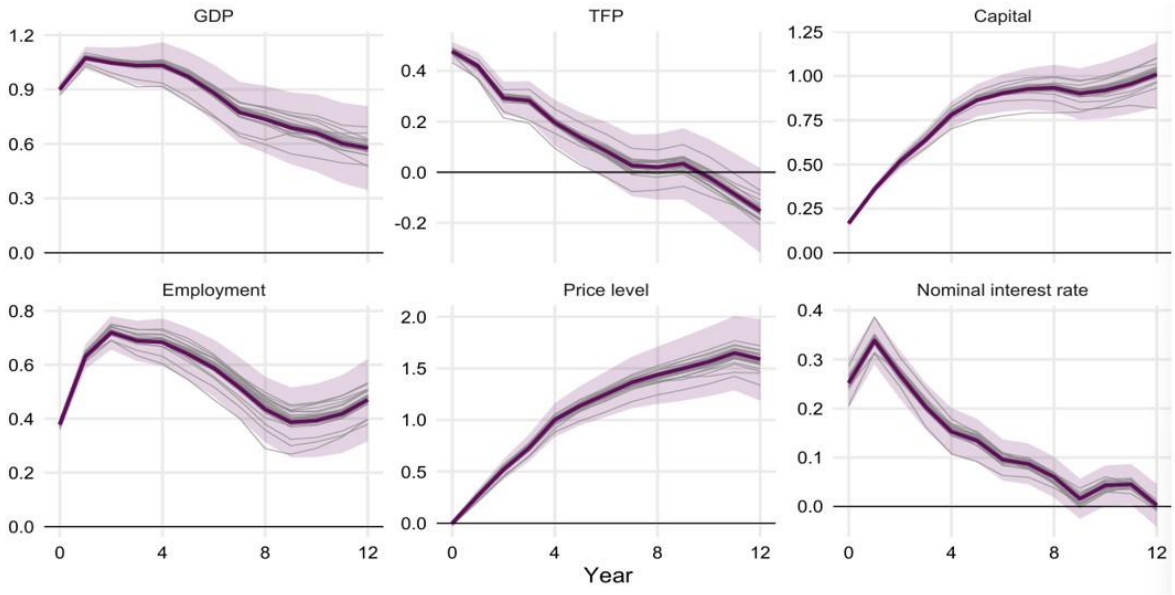
Source: Own calculations using data from Müller et al. (2025). Note: In all panels, teal dots represent countries where the posterior mean is included in both posterior 90% CI, purple if this only happens once, and red if both posterior 90% CI exclude the posterior mean.

G Robustness checks of the panel local projections

This appendix examines the robustness of the pooled local projections of Section 5.

The first check re-estimates the projections excluding one country at a time and overlays the thirty resulting median IRFs on the baseline band (Figure G1). This matters because the panel pools economies of very different size and cyclical history, so the estimated persistence could in principle be the imprint of a single influential country (a southern European crisis economy, say) through the severity of its episodes rather than a property of the cross-country data. The result is that no exclusion alters the pattern: the thirty leave-one-out paths remain within, or marginally outside, the baseline 68 percent band at every horizon, the sign of the long-horizon employment, capital, and GDP responses is preserved in all thirty cases, and the most influential exclusions — Japan for capital (1.01 to 0.82 at horizon twelve) and Spain for employment (0.47 to 0.38) — leave the estimates well away from zero.

Figure G1: Hairplots dropping each country one at a time

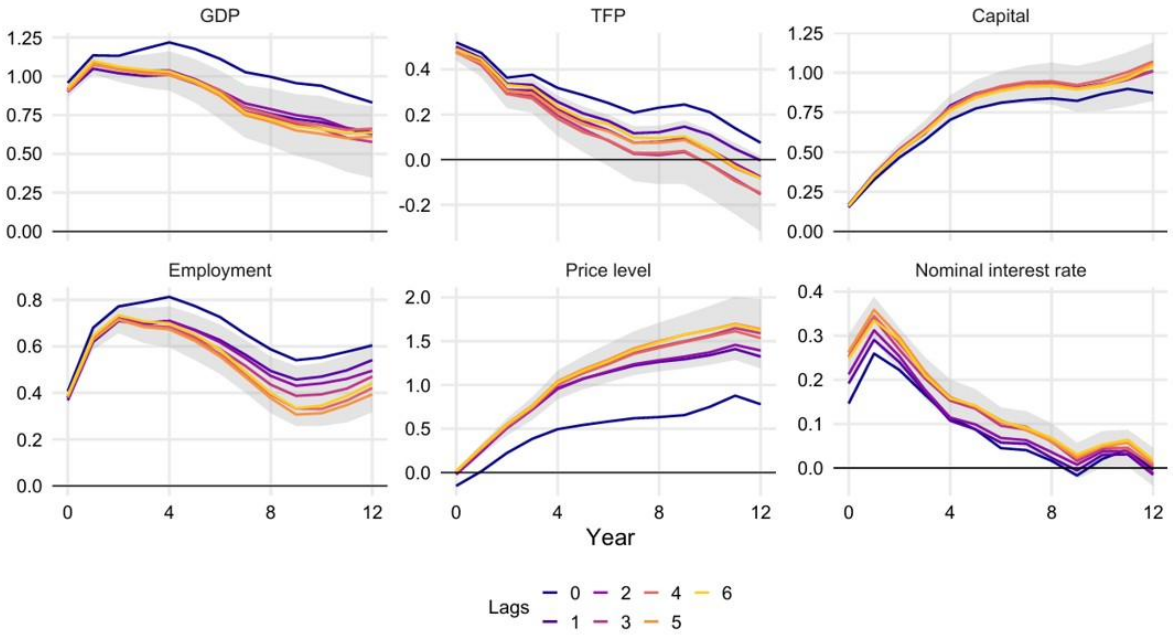


Source: Own calculations using the posterior distributions from the estimates computed with data from Müller et al. (2025).

The second check varies the number of lags of the differenced outcome from zero to six (Figure G2). This matters because the dynamic specification of a local projection is chosen by convention rather than derived from theory, and estimated persistence can be sensitive to how much of the outcome's own history is controlled for. The result is stability from one lag onward: across lag orders the horizon-twelve employment median

ranges from 0.39 to 0.60 and the capital median from 0.87 to 1.07. The specification with no lags differs modestly — output and TFP responses shift up and the price response down, as expected when predictable outcome dynamics are left uncontrolled — and is reported for completeness.

Figure G2: Sensitivity analysis to the inclusion of lags of the dependent variable first differences

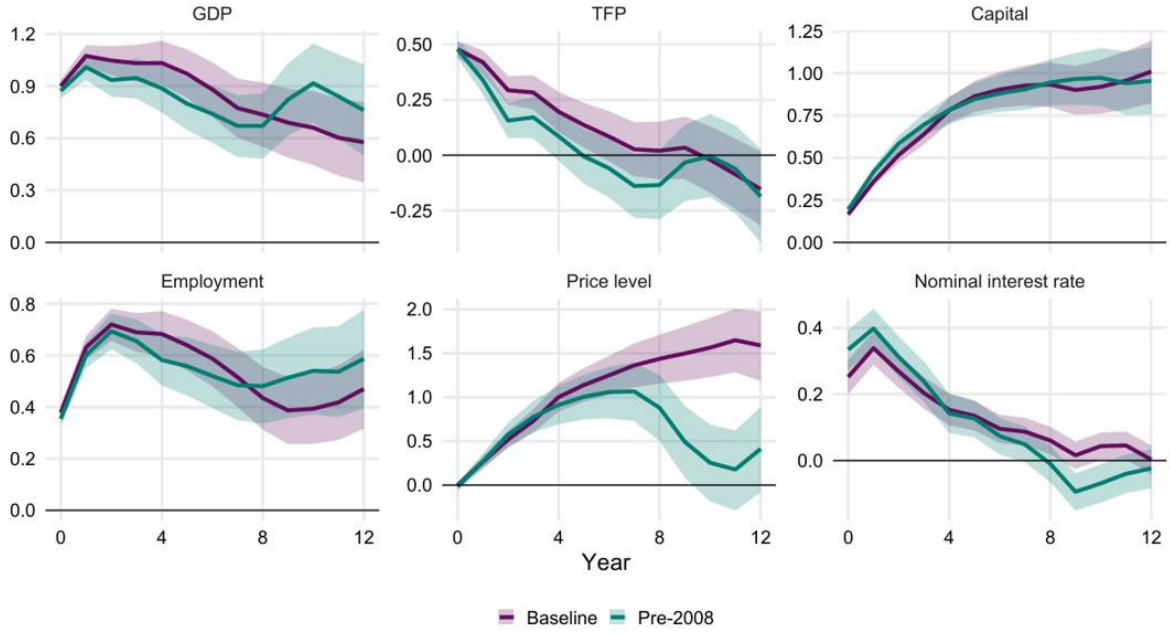


Source: Own calculations using the posterior distributions from the estimates computed with data from Müller et al. (2025).

The third check truncates the panel in 2007, excluding both the global financial crisis and the pandemic (Figure G3). This matters because these two episodes dominate recent macroeconomic variation, and a natural concern is that the estimated hysteresis merely reflects them — that the persistence documented in Section 5 is the signature of 2008 and 2020 rather than a general feature of demand shocks. The result is the opposite: the pattern is, if anything, stronger in the earlier sample, with the horizon-twelve employment median rising to 0.59 ([0.40, 0.78]) and the GDP median to 0.76 ([0.50, 1.02]), while the capital response is unchanged. The one exhibit that is sample-dependent is the price level, whose long-run response falls from 1.59 ([1.19, 1.98]) in the full sample to 0.41 ([–0.08, 0.88]) before 2008, indicating that the strong long-run price adjustment is concentrated in the post-2008 period.

The fourth check adds two lags of the shock draw itself to the projection (Figure G4). This matters because smoothed shock estimates are serially correlated by construction —

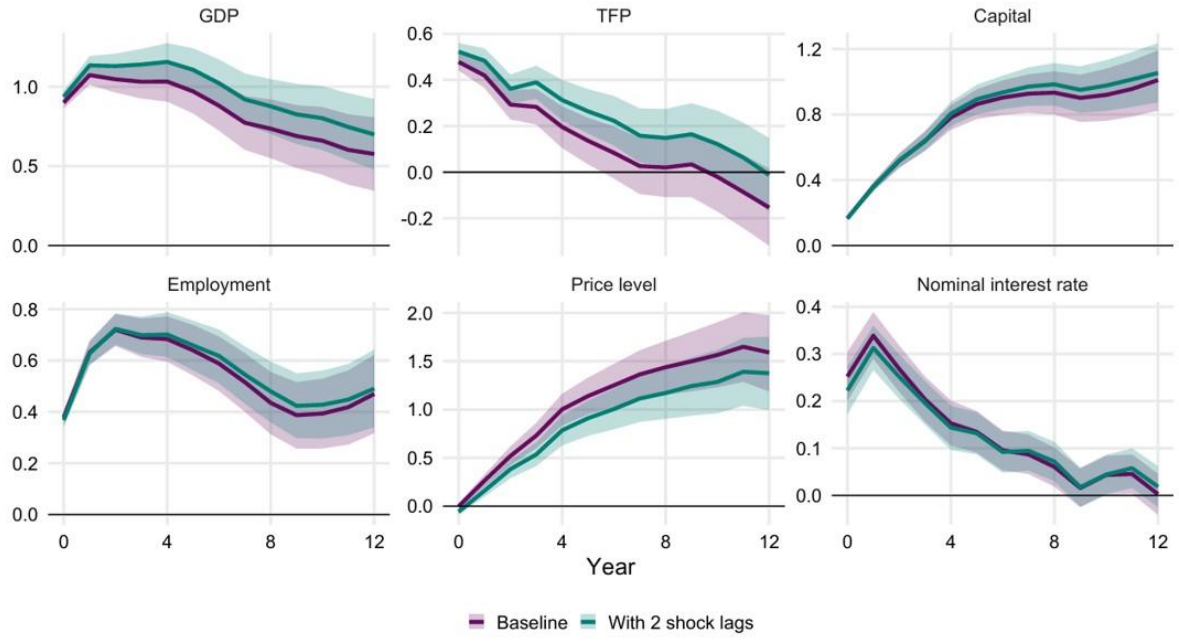
Figure G3: Influence of the sample period on the local projections



Source: Own calculations using the posterior distributions from the estimates computed with data from Müller et al. (2025).

the smoother borrows information from neighboring periods — so an estimated impulse response can partly trace the persistence of the shock measure rather than the persistence of the response. The result is that the employment and capital responses are essentially unchanged (0.49 and 1.05 at horizon twelve), confirming that their persistence belongs to the response and not to the shock measure, while the long-run TFP estimate moves from –0.15 to –0.01.

Figure G4: Robustness of the local projections to lags of the shock



Source: Own calculations using the posterior distributions from the estimates computed with data from Müller et al. (2025).

H Moderator data

Table H1: Sources and windows of the fifteen moderators

Moderator	Source	Window
Investment share of GDP	Penn World Table	1980–2019
Government share of GDP	Penn World Table	1980–2019
Labour share	Penn World Table	1980–2019
Population aged 65+	WDI	1980–2019
CPI inflation	WDI	1987–2019
Self-employment	WDI	1991–2019
Trade openness	WDI	1991–2019
Manufacturing % of value added	WDI	2004–2019
Employment protection	OECD, version 1	2008–2019[a]
Private credit % of GDP	WDI	2001–2010
Product market regulation	OECD, 2023 methodology	2018 vintage
Union density	OECD and academic research	2010–2019
Bargaining coverage	ILO	2000–2019[b]
R&D % of GDP	WDI	1996–2019[b]
Years at the ELB, share	Own construction, BIS policy rates	1980–2019[c]

Notes: The first ten are averaged over a window common to all economies; the last five are cross-sections. [a] China and South Africa enter on 2008–2012 and Colombia on 2014–2019. [b] Available years within the window; Israel's bargaining coverage rests on two observations and Switzerland's R&D on eight. [c] Euro-area members share the ECB's value.

Source: Own elaboration.